

Alteration of long and short-term hematopoietic stem cell ratio causes myeloid-biased hematopoiesis

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Katsuyuki Nishi, Taro Sakamaki, Akiomi Nagasaka, Kevin S Kao, Kay Sadaoka, Masahide Asano, Nobuyuki Yamamoto, Akifumi Takaori-Kondo, Masanori Miyanishi 

Hematopoietic Stem Cell Biology and Medical Innovation (HSCBMI), Department of Pediatrics, Kobe University Graduate School of Medicine, Kobe, Japan • RIKEN Center for Biosystems Dynamics Research, Kobe, Japan • Weill Cornell, Rockefeller, Sloan-Kettering, Tri-Institutional MD-PhD Program, New York, United States • Institute of Laboratory Animals, Kyoto University Graduate School of Medicine, Kyoto, Japan • Department of Pediatrics, Kobe University Graduate School of Medicine, Kobe, Japan • Department of Hematology and Oncology, Kyoto University Graduate School of Medicine, Kyoto, Japan

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eLife Assessment

This manuscript provides **valuable** insights into the heterogeneity of hematopoietic stem cells and age-associated myeloid-biased hematopoiesis. While several aspects of the study are intriguing and merit further investigation, the current results remain **incomplete** and additional data are necessary to substantiate the conclusions. Some of the methods and data analyses partially support the claims.

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Abstract

Myeloid-biased hematopoiesis is a well-known age-related alteration. Several possibilities, including myeloid-biased hematopoietic stem cell (HSC) clones, may explain this. However, the precise mechanisms remain controversial.

Utilizing the Hoxb5 reporter system to prospectively isolate long-term HSCs (LT-HSCs) and short-term HSCs (ST-HSCs), we found that young and aged LT-HSCs co-transplanted into the same recipients demonstrated nearly equivalent myeloid lineage output, contrary to the theory of myeloid-biased HSC clones. Transcriptomics indicated no significant myeloid gene enrichment in aged LT-HSCs compared to their young counterparts. Instead, transplanting reconstituted young HSCs with the ratio of LT/ST-HSCs seen in aged mice can significantly skew the lineage output to myeloid cells. In addition, while the niche environment in the bone marrow minimally affects myeloid-biased hematopoiesis, aged thymi and spleens substantially hinder lymphoid hematopoiesis, resulting in further myeloid-domination. Thus, we demonstrate that myeloid-biased hematopoiesis in aged organisms originates due to alteration of the ratio between LT-HSCs and ST-HSCs rather than in heterogeneous HSC clones with various cell fates.

Introduction

Age-associated changes in individuals are deeply correlated with progressive attenuation of cellular functions in tissue stem cells of organs (Jones and Rando 2011 [↗](#)). In the hematopoietic system, hematopoietic stem cells (HSCs), which possess self-renewal capacity and multipotency (Spangrude et al., 1988 [↗](#); Weissman and Shizuru 2008 [↗](#); Majeti et al., 2007 [↗](#)), are responsible for various hematopoietic alterations with age, such as reduced self-renewal capacity (Morrison et al., 1996 [↗](#)) and myeloid-biased hematopoiesis (Muller-Sieburg et al., 2004 [↗](#); Challen et al., 2010 [↗](#)) due to their functional decline. For example, myeloid-biased hematopoiesis potentially reduces response to infections (Webster, 2000 [↗](#)), reduces vaccination efficacy (Crooke et al., 2019 [↗](#)), and increases myeloid malignancy (Rossi et al., 2008 [↗](#)) in aged individuals. To comprehend these age-associated physiological changes, myeloid-biased hematopoiesis has been studied in considerable detail.

Based on the experimental observation that transplantation of aged HSCs exhibits more myeloid-biased differentiation in young recipient mice than transplantation of young HSCs, this phenotype has been thought to originate in cell-intrinsic changes in the HSC compartment (Geiger et al., 2013 [↗](#); De Haan and Lazare 2018 [↗](#); Schultz and Sinclair 2016 [↗](#)). A myeloid-biased phenotype after aged HSC transplantation leads to the suggestion that myeloid-biased HSC clones selectively expand with age (Cho et al., 2008 [↗](#); Pang et al., 2011 [↗](#)). Transcriptome and epigenetic analyses showing that a set of genes related to myeloid differentiation is significantly enriched in aged HSCs compared to young HSCs (Rossi et al., 2005 [↗](#); Grover et al., 2016 [↗](#)) further supports this hypothesis.

On the other hand, some reports support a different point of view in light of experimental evidence showing that aged lymphoid-biased HSCs still demonstrate the same level of lymphopoiesis as their younger counterparts, despite exhibiting a myeloid-biased gene expression pattern (Montecino-Rodriguez et al., 2019 [↗](#)). This result highlights the limitations of predicting a pattern of HSC differentiation based upon gene expression patterns. Moreover, myeloid progenitors such as granulocyte-macrophage progenitor (GMP) and common myeloid progenitor (CMP) should also increase with age if the selective expansion of myeloid-biased HSCs leads to an increase of myeloid cells in peripheral blood (PB). However, increment of such progenitors with age was not consistently demonstrated in earlier research (Rossi et al., 2005 [↗](#); Nilsson et al., 2016 [↗](#); Min et al., 2006 [↗](#)). Additionally, a mathematical model demonstrated that aging had no effect on the daily production of CMP supplied from multipotent progenitors (MPP) (Busch et al., 2015 [↗](#); Dorshkind et al., 2020 [↗](#)).

As such, while selective expansion of myeloid-biased HSC clones is the most widely accepted hypothesis to explain myeloid-biased hematopoiesis in aging (Mejia-Ramirez and Florian, 2020 [↗](#); SanMiguel et al., 2020 [↗](#)), other mechanisms may exist in light of the inconsistent cellular behavior of progenitor fractions relative to HSCs. To the best of our knowledge, no reports have analyzed kinetics of age-associated changes in HSCs, progenitors, and PB cells simultaneously. Therefore, we started our investigation by examining the correlation between age-related changes in PB and BM to shed light on the mechanism underlying myeloid-biased hematopoiesis that occurs during aging.

Results

The discrepancy of age-associated alternation in peripheral blood and bone marrow calls into question the existence of a myeloid-biased clone

Mouse HSC research on aging has used mice aged 18 months or older (Challen et al., 2010 [↗](#); Cho et al., 2008 [↗](#); D. J. Rossi et al., 2005 [↗](#); Grover et al., 2016 [↗](#); Montecino-Rodriguez et al., 2019 [↗](#); Nilsson et al., 2016 [↗](#)). However, given that continuous accumulation of cellular stress with age causes a gradual decline of cellular functions, a comprehensive analysis from young to old mice is necessary to unravel mechanisms by which age-associated, myeloid-biased hematopoiesis progress. Hence, we analyzed changes in PB (Fig. S1 A) at multiple time points from young to old mice. These results showed that the percentage of myeloid cells began to change as early as 6 months in mice and continued to increase at a constant rate until the age of ≥ 23 months (Fig. 1 A [↗](#)).

It has been reported that myeloid-biased hematopoiesis is caused by a selective increase of myeloid-biased clones in the immunophenotypically defined (surface-antigen defined) HSC fraction (Fig. S1 B) (Beerman et al., 2010 [↗](#)). According to this hypothesis, age-associated myeloid hematopoiesis progression in PB would be paralleled by an increase in myeloid-biased HSC clones. Therefore, we examined the frequency of the HSC and progenitor fractions in the BM at multiple time points (Fig. S1, B and C). We found that the frequency of immunophenotypically defined HSC in BM rapidly increased up to the age of 12 months. After the age, the rate of increase in their frequency appeared to slow down. (Fig. 1 B [↗](#)). On the other hand, in contrast to what we anticipated, the frequency of GMP was stable, and the percentage of CMP actually decreased significantly with age, defying our prediction that the frequency of components of the myeloid differentiation pathway, such as CMP, GMP, and MEP would increase in aged mice if myeloid-biased HSC clones increase with age (Fig. 1 B [↗](#)).

Finally, we determined the ratio of each fraction in young mice versus aged mice to compare the age-associated transition pattern of components comprising the myeloid differentiation pathway. Our analysis of hematopoietic alternations with age revealed that age-associated transition patterns of immunophenotypically defined HSC and CMP in BM were not paralleled with myeloid cell in PB (Fig. 1 C [↗](#)). These findings called into question the hypothesis that there is a selective increase of myeloid-biased HSC clones in the aged BM. We then set out to elucidate the mechanism by which myeloid-biased phenotypes arise in the PB of aged mice.

The long-term observation of 2-year-old LT-HSC's differentiation does not indicate the expansion of myeloid-biased clones

Numerous studies have claimed that the myeloid-skewed phenotype observed in PB of aged mice is caused by myeloid-biased HSC clones selectively expanded from HSCs with originally heterogeneous differentiation potentials (Beerman et al., 2010 [↗](#); Dykstra et al., 2011 [↗](#); Sudo et al., 2000 [↗](#)). Other studies have reported that immunophenotypically defined HSCs possess heterogeneity associated with self-renewal capacity, suggesting the existence of at least two different populations in the HSC compartment, LT-HSCs and ST-HSCs (Morrison and Weissman 1994 [↗](#); Spangrude et al., 1995 [↗](#)). Since LT-HSCs have extensive self-renewal capacity, while ST-HSCs lose their self-renewal capacity within a short period, LT-HSCs are thought to persist in the BM throughout life and to enrich age-associated changes compared to ST-HSCs. Therefore, we expect that LT-HSC-specific analyses will help to answer whether myeloid-biased HSC clones exist.

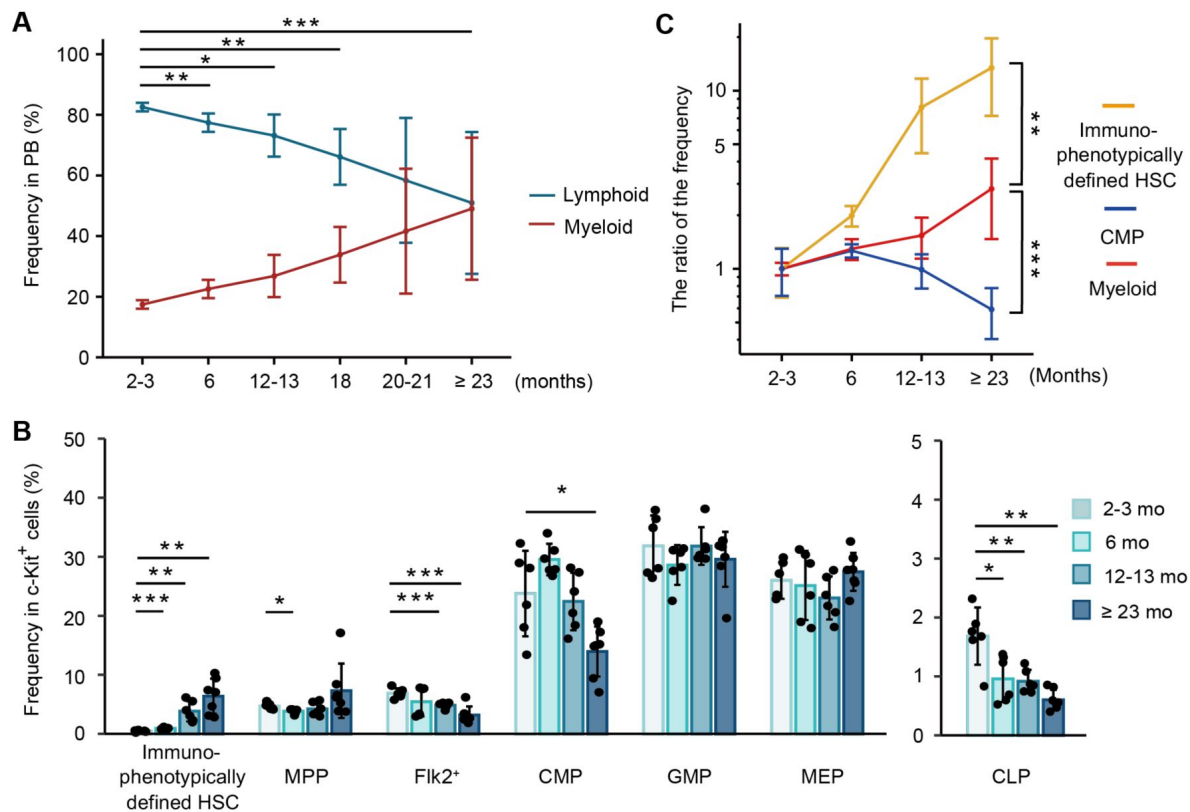


Figure 1.

Comprehensive analysis of hematopoietic alternations with age shows a discrepancy of age-associated changes between peripheral blood and bone marrow.

(A) Average frequency of myeloid cells (neutrophils, monocytes) and lymphoid cells (B cells, T cells, and NK cells) in PB at the age of 2-3 months ($n = 6$), 6 months ($n = 6$), 12-13 months ($n = 6$), 18 months ($n = 6$), 20-21 months ($n = 5$), ≥ 23 months ($n = 16$). Abbreviation: PB = Peripheral blood.

(B) Average frequency of immunophenotypically defined HSC and progenitor cells in BM of 2-3-month mice ($n = 6$), 6-month mice ($n = 6$), 12-13-month mice ($n = 6$), ≥ 23 -month mice ($n = 7$).

(C) Age-associated changes of immunophenotypically defined HSC and myeloid differentiation components (CMP and myeloid cells in the PB). The ratio of aged to young frequency was calculated as (the fraction frequency at each aged mice (%)) / (the average fraction frequency at 2-3-month mice (%)). $*P < 0.05$. $**P < 0.01$. $***P < 0.001$. Data and error bars represent means $\pm$ standard deviation.

We previously reported that *Hoxb5* exclusively marks LT-HSCs in young mice (Chen et al., 2016 [DOI](#)). However, we have not tested in aged mice whether *Hoxb5* specifically labels LT-HSCs and helps to distinguish LT-HSCs and ST-HSCs from immunophenotypically defined HSCs (hereafter, bulk-HSCs). To confirm this, we first analyzed expression of *Hoxb5* in bulk-HSC, MPP, Flk2^+ , and $\text{Lin}^- \text{Sca1}^c\text{-Kit}^+$ populations in aged *Hoxb5*-tri-mCherry mice. We observed that *Hoxb5* was exclusively expressed in bulk-HSCs (Fig. 2 A [DOI](#)). Then, to verify long-term engraftment, we conducted a transplantation assay utilizing *Hoxb5*⁺ HSCs and *Hoxb5*⁻ HSCs, respectively, isolated from 2-year-old mice (Fig. 2 B [DOI](#)). We observed that only recipients receiving aged *Hoxb5*⁺ HSCs exhibited continuous hematopoiesis 16 weeks after primary transplantation (Fig. 2, C and E [DOI](#); Fig. S2 A). In secondary transplantation analysis, only recipients receiving *Hoxb5*⁺ HSCs exhibited robust hematopoiesis throughout the period of observation, indicating that *Hoxb5* can be used as a specific marker of LT-HSCs in aged mice, as well as young mice (Fig. 2, D and F [DOI](#); Fig. S2 B).

A serial transplantation assay with a period of observation longer than 8 months should be long enough to observe further myeloid-biased change. If bulk-HSCs isolated from aged mice are already enriched by myeloid-biased HSC clones, we should see more myeloid-biased phenotypes 16 weeks after primary and the secondary transplantation. However, we found that kinetics of the proportion of myeloid cells in PB were similar across the primary and the secondary transplantation (Fig. 2 G [DOI](#)). These results suggest the following two possibilities: either myeloid-biased HSCs do not expand in the LT-HSC fraction, or the expansion of myeloid-biased clones in 2-year-old mice has already peaked.

Direct comparison between young vs. aged LT-HSC differentiation reveals that LT-HSCs exhibit unchanged lineage output throughout life

We developed a co-transplantation assay to directly compare the differentiation capacity between young and aged LT-HSCs by co-transplanting ten young GFP⁺ LT-HSCs and ten aged GFP⁻ LT-HSCs into the same recipient mice (Fig. 3 A [DOI](#)). Then, we found that the myeloid lineage proportions from young and aged LT-HSCs were nearly comparable during the observation period after transplantation (Fig. 3, B and C [DOI](#)). Furthermore, by analyzing the proportion of mature cell types derived from young and aged LT-HSCs in the same donor, we directly compared the capacity for hematopoietic reconstitution in each mature cell type between young and aged LT-HSCs. We confirmed again that the reconstitution ratio was almost the same across all lineages, although bulk hematopoiesis derived from young LT-HSCs predominated (Fig. 3 D [DOI](#)). These results indicate that the differentiation potential of LT-HSCs remains unchanged throughout their lives.

Several reports have demonstrated that transplanting mixed/bulk-HSCs—a combination of LT-HSCs and ST-HSCs—obtained from old animals results in blatantly myeloid-biased hematopoiesis (Beerman et al., 2010 [DOI](#); Dykstra et al., 2011 [DOI](#); Sudo et al., 2000 [DOI](#)). To test this, we co-transplanted bulk-HSCs from young and old mice to corroborate this (Fig. S3 A). As previously described, we observed that aged bulk-HSC exhibited a myeloid-skewed phenotype (Fig. S3 B and C). Additionally, we observed that aged bulk-HSC reconstitution exhibited higher reconstitution of myeloid cells compared to young bulk-HSCs (Fig. S3 D). These findings unmistakably demonstrated that mixed/bulk-HSCs showed myeloid skewed hematopoiesis in PB with aging, while LT-HSCs exhibited unchanged lineage output throughout life.

LT-HSCs never show myeloid-related gene set enrichment during aging

Although the results of co-transplantation of young and aged LT-HSCs did not demonstrate lineage-biased output by LT-HSCs throughout life, enrichment of myeloid genes in aged bulk-HSCs shown by previous studies supports the idea that myeloid-biased hematopoiesis is caused by selective

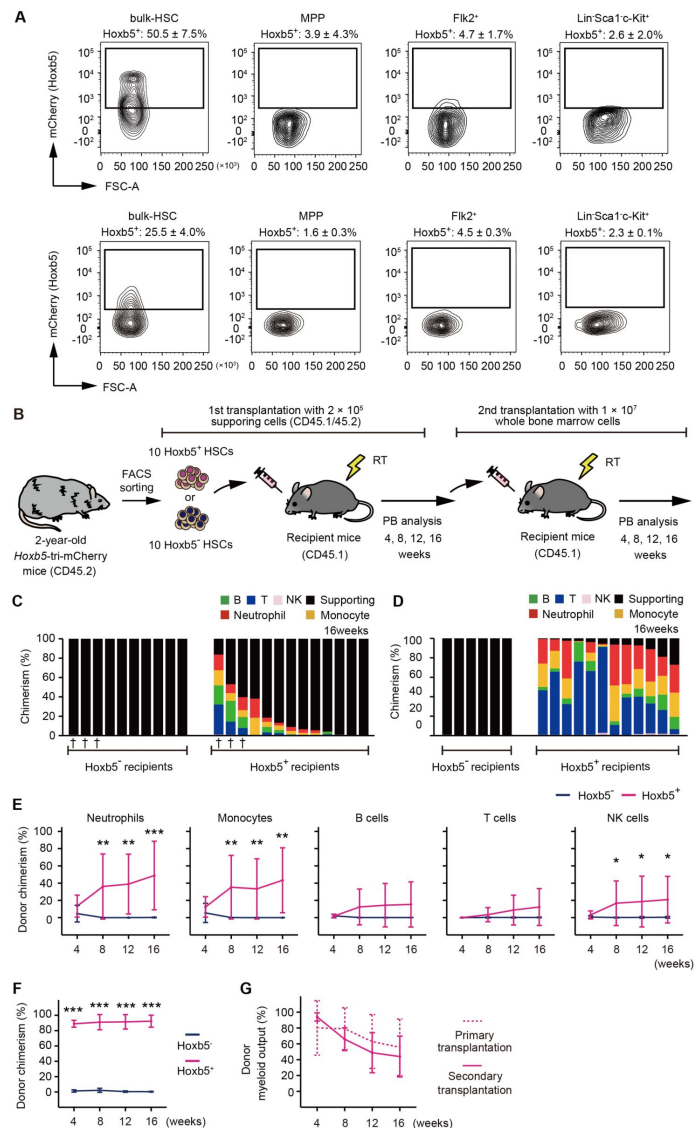


Figure 2.

The expansion of myeloid-biased clones was not observed in 2-year-old LT-HSCs after their transplantation.

(A) *Hoxb5* reporter expression in bulk-HSC, MPP, Flk2⁺, and Lin⁻Sca1⁺c-Kit⁺ populations in the 2-year-old *Hoxb5*-tri-mCherry mice (Upper panel) and 3-month-old *Hoxb5*-tri-mCherry mice (Lower panel). Values indicate the percentage of mCherry⁺ cells $\pm$ standard deviation in each fraction ($n = 3$).

(B) Experimental design to assess the long-term reconstitution ability of *Hoxb5*⁺ HSCs or *Hoxb5*⁻ HSCs. *Hoxb5*⁺ HSCs and *Hoxb5*⁻ HSCs were isolated from 2-year-old CD45.2 *Hoxb5*-tri-mCherry mice and were transplanted into lethally irradiated CD45.1 recipient mice with 2×10^5 supporting cells (*Hoxb5*⁺ HSCs, $n = 13$; *Hoxb5*⁻ HSCs, $n = 10$). For secondary transplants, 1×10^7 whole BM cells were transferred from primary recipient mice. Abbreviations: PB = Peripheral blood, RT = Radiation therapy.

(C) Percentage chimerism at 16 weeks after receiving ten aged *Hoxb5*⁺ HSCs or ten aged *Hoxb5*⁻ HSCs. Each column represents an individual mouse.

(D) Percentage chimerism at 16 weeks after whole BM secondary transplantation. Donor whole BM cells for secondary transplantation were taken from mice denoted by † in **Figure 2C**.

(E) Kinetics of average donor chimerism in each PB fraction after primary transplantation.

(F) Kinetics of average donor chimerism after secondary transplantation.

(G) Kinetics of average donor myeloid output (myeloid proportion in donor cells) in LT-HSC recipient mice after primary and secondary transplantation. * $P < 0.05$. ** $P < 0.01$. *** $P < 0.001$. Data and error bars represent means $\pm$ standard deviation.

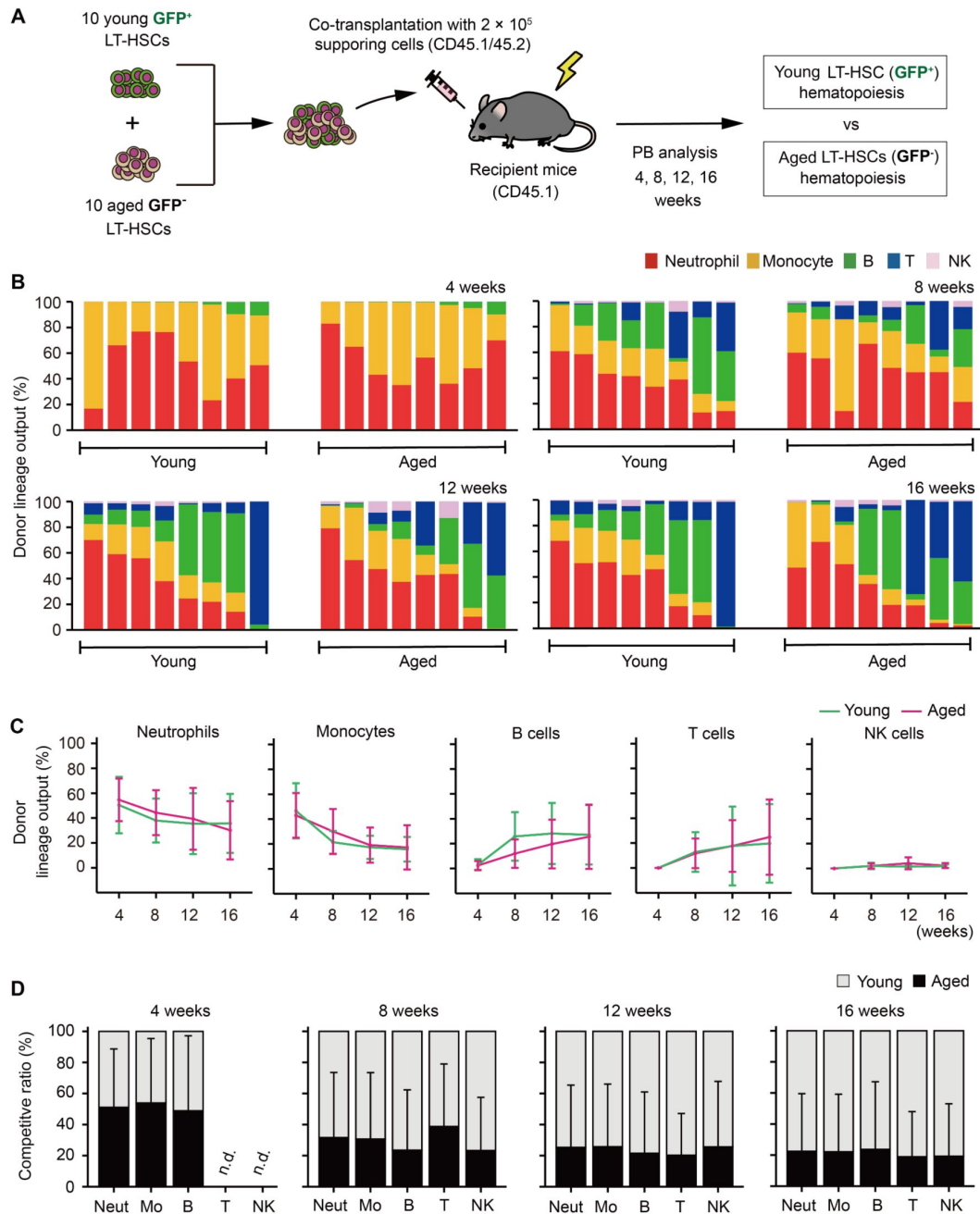


Figure 3.

Aged LT-HSCs show balanced hematopoiesis throughout life.

(A) Experimental design for competitive co-transplantation assay using young LT-HSCs sorted from *Hoxb5*-tri-mCherry GFP mice and aged LT-HSCs sorted from *Hoxb5*-tri-mCherry mice. Ten CD45.2⁺ young LT-HSCs and ten CD45.2⁺ aged LT-HSCs were transplanted with 2×10^5 CD45.1⁺/CD45.2⁺ supporting cells into lethally irradiated CD45.1⁺ recipient mice ($n = 8$).

(B) Lineage output of young or aged LT-HSCs at 4, 8, 12, 16 weeks after transplantation. Each bar represents an individual mouse.

(C) Lineage output kinetics of young LT-HSCs or aged LT-HSCs at 4, 8, 12, 16 weeks post-transplant.

(D) Competitive analysis of young LT-HSCs vs. aged LT-HSCs lineage output at 4, 8, 12, 16 weeks post-transplant. Competitive ratio was calculated as the proportion of young LT-HSCs derived cells vs. aged LT-HSCs derived cells in each fraction.

Abbreviations: Neut = Neutrophils, Mo = Monocytes, B = B cells, T = T cells, and NK = NK cells. Data and error bars represent means $\pm$ standard deviation. "n.d." stands for "not detected."

expansion of myeloid-biased HSC clones (D. J. Rossi et al., 2005 [DOI](#); Grover et al., 2016 [DOI](#)). To compare age-associated myeloid gene enrichment, we isolated bulk-HSC, LT-HSCs, and ST-HSCs from young or aged mice (**Fig. 4 A** [DOI](#)).

To ensure the quality of the sorted fraction for subsequent RNA-seq analyses, we verified the *Hoxb5* read counts (Fig. S4 A). Cluster dendrograms using the whole transcriptome confirmed that cell fractions isolated from young and aged mice clustered into distinct groups (**Fig. 4 B** [DOI](#)). Then, using sets of aging-related genes, such as inflammation (Liberzon et al., 2015 [DOI](#); Pietras 2017 [DOI](#)), DNA damage (p53 pathway) (Liberzon et al., 2015 [DOI](#); D. J. Rossi et al., 2007 [DOI](#)), cell cycle progression (E2F target) (Liberzon et al., 2015 [DOI](#); Kowalczyk et al., 2015 [DOI](#)), and a common aging signature (Flohr Svendsen et al., 2021 [DOI](#)), we ran a violin plot analysis on each cell fraction to confirm the occurrence of typical aging-related gene expression changes. We found age-associated changes in inflammation and the DNA damage between cells isolated from young and aged mice, but very similar patterns regardless of cell fraction in young or aged mice. In terms of cell cycle progression and aging signature genes, however, we discovered that only the young LT-HSC fraction differed from other young fractions and tended to represent similar gene expression pattern with aged fractions (**Fig. 4 C** [DOI](#)). Then, taking LT-HSC specific gene expression pattern into account, we looked at expression patterns of specific genes that had been highly validated by experiments as being associated with myeloid-biased HSC (Beerman et al., 2010 [DOI](#); Flohr Svendsen et al., 2021 [DOI](#); Sanjuan-Pla et al., 2013 [DOI](#); Mann et al., 2018 [DOI](#)). We discovered that expression of these genes is relatively comparable between young and aged LT-HSCs, but not other fractions (Fig. S4 B).

A recent comprehensive analysis of mouse HSC aging using multiple RNAseq data sets claimed that almost 80% of differentially expressed genes are poorly reproducible across data sets (Flohr Svendsen et al., 2021 [DOI](#)). In fact, we found that almost 80% of genes are not shared when we compare three representative myeloid/lymphoid gene sets (Sanjuan-Pla et al., 2013 [DOI](#); Chambers et al., 2007 [DOI](#); Pronk et al., 2007 [DOI](#)). Additionally, most genes were only used in a single gene set (**Fig. 4, D and E** [DOI](#)). By using only genes that were shared in these gene sets, we ran a gene set enrichment analysis (GSEA) to see whether myeloid/lymphoid genes were enriched in aged LT-HSCs or other fractions. Neither aged LT-HSCs nor aged ST-HSCs exhibited myeloid/lymphoid gene set enrichment, while shared myeloid-related genes tended to be enriched in aged bulk-HSCs, although this enrichment was not statistically significant (**Fig. 4, F and G** [DOI](#)). On the other hand, GSEA analysis using original gene sets, respectively, showed inconsistent results (Fig. S4, C and D).

A myeloid-biased phenotype in peripheral blood depends on the relative decrease of ST-HSC in the HSC compartment with age

While transplantation of aged bulk-HSCs exhibits a myeloid-biased phenotype in PB shown in previous reports (Beerman et al., 2010 [DOI](#); Dykstra et al., 2011 [DOI](#); Sudo et al., 2000 [DOI](#)) and our results (Fig. S3), aged LT-HSCs remain stable in terms of the balance of their differentiation potential between myeloid and lymphoid production (**Fig. 3** [DOI](#)). Comparative transcriptome analyses demonstrated that neither LT-HSCs nor ST-HSCs, which are functionally more homogeneous than bulk-HSCs, exhibited myeloid gene set enrichment with age, whereas aged bulk-HSCs tended to show more myeloid gene set enrichment than their young counterparts (**Fig. 4** [DOI](#)). We then tried to figure out what causes the myeloid-biased phenotype in PB after transplantation of bulk-HSCs.

Because LT-HSCs in bulk-HSCs exhibit nearly constant level of hematopoiesis throughout life, we hypothesized that ST-HSCs could be the key to discovering mechanisms underlying the myeloid-skewing phenomenon. First, we revisited our previous study (Chen et al., 2016 [DOI](#)), which demonstrated that transplanted ST-HSCs maintain lymphocyte production while rapidly losing myeloid lineage production in recipients. This result may potentially indicate the presence of HSC clones with a preference for lymphoid differentiation. Given that donor cells obtained from

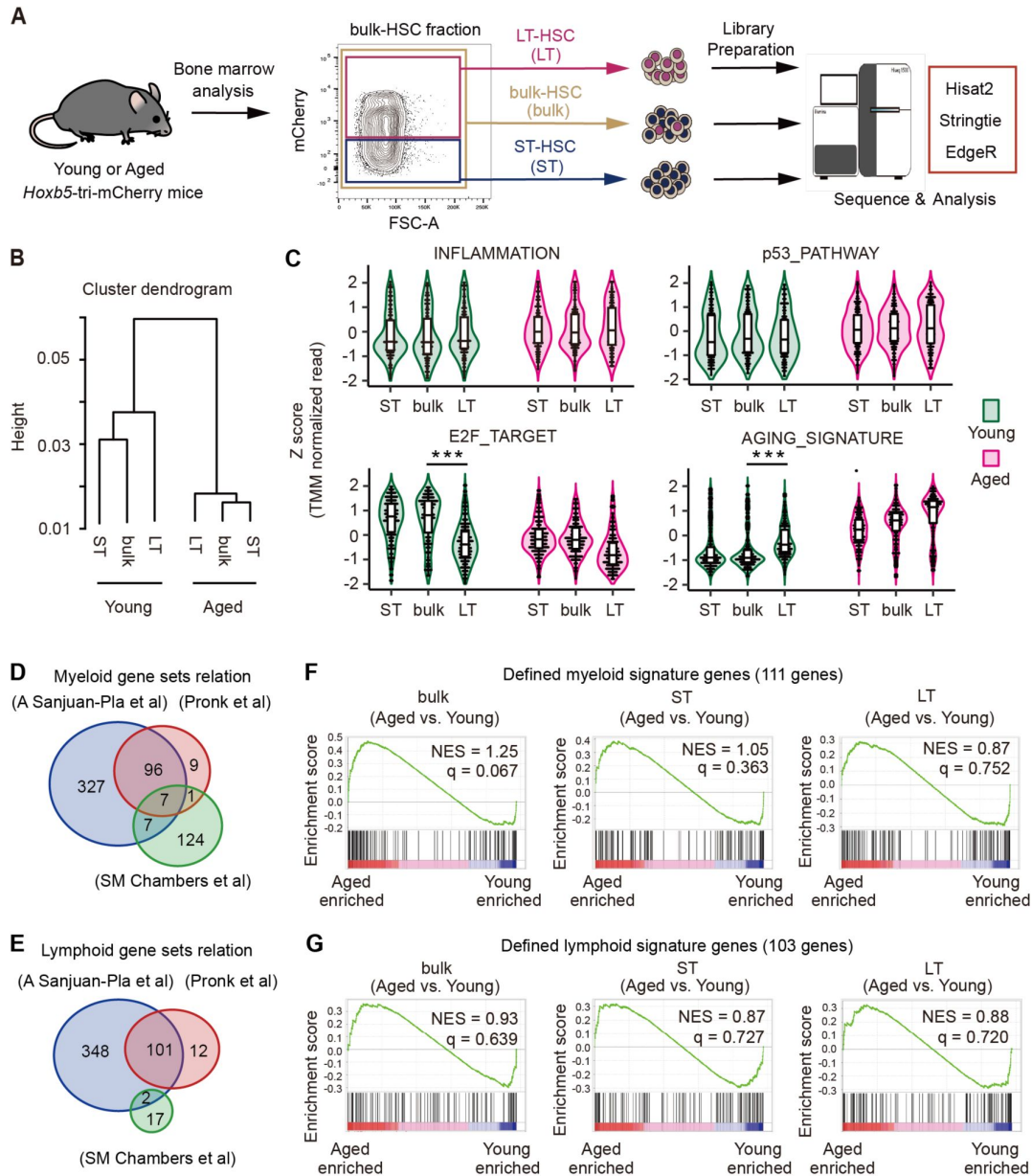


Figure 4.

Myeloid-associated genes were not enriched in aged LT-HSCs compared to their young counterparts.

(A) Experimental schematic for transcriptome analysis. LT-HSCs ($n = 3$), ST-HSCs ($n = 3$), and bulk-HSCs ($n = 3$) were sorted from young (2-3 months) or aged (23-25 months) *Hoxb5*-tri-mCherry mice, after which each RNA was harvested for RNA sequencing.

(B) Hierarchical clustering dendrogram of whole transcriptomes using Spearman distance and the Ward clustering algorithm. (C) Violin plots showing normalized gene expression levels for each gene set in young and aged LT-HSCs, ST-HSCs, and bulk-HSCs. Expression values for each gene were standardized independently by applying Z score transformation.

(D, E) Venn diagram showing the overlap of genes between three myeloid signature gene sets, and lymphoid signature gene sets (A Sanjuan-Pla et al., 2013 [\[1\]](#), Pronk et al., 2007 [\[2\]](#), SM Chambers et al., 2007 [\[3\]](#)).

(F, G) Signature enrichment plots from GSEA analysis using defined myeloid and lymphoid signature gene sets that overlapped in the three gene sets. Values indicated on individual plots are the normalized Enrichment Score (NES) and q -value of enrichment. *** $P < 0.001$.

primary recipients receiving ST-HSCs do not undergo hematopoiesis in secondary transplantation (Chen et al., 2016), it is likely that all ST-HSCs have already lost their ability to self-renew and have disappeared from the HSC fraction in primary recipients. To confirm this, we transplanted ten LT-HSCs or ST-HSCs and then analyzed PB and BM (Fig. 5 A). As we previously reported (Chen et al., 2016), donor-derived hematopoiesis eventually becomes lymphocyte-dominant in PB of ST-HSC recipients over time (Fig. 5 B). Subsequently, it was determined that the vast majority of T cell subsets are memory cells (Fig. 5, C and D). In addition, unlike recipients receiving LT-HSCs, recipients receiving ST-HSCs lacked any bulk-HSCs in their BM (Fig. 5, E and F). These results strongly suggest that lymphoid-biased hematopoiesis observed after transplantation of ST-HSCs is due to persistence of memory-type lymphocytes in PB, rather than to *de novo* lymphopoiesis from lymphoid-biased HSC clones. Persistence of memory-type lymphocytes from ST-HSCs may have led us to misinterpret mechanisms underlying myeloid-biased hematopoiesis after aged bulk-HSC transplantation. To verify this hypothesis, we examined kinetics of the LT-HSC/ST-HSC ratio in the bulk-HSC population, revealing the ratio of ST-HSC to LT-HSC decreased with age (Figure S5). Thus, we hypothesized that the relative decrease in the ST-HSC ratio in the aged bulk-HSC fraction would lead to reduction of memory-type lymphocytes and myeloid-biased hematopoiesis following transplantation of aged bulk-HSCs. To test this hypothesis, we isolated LT-HSCs and ST-HSCs from young donor mice and reconstituted them with a 2:8 ratio (as in young mice) or a 5:5 ratio (as in aged mice) prior to transplanting them (Fig. 6 A). Four weeks after transplantation, PB analysis revealed that both groups had comparable patterns of hematopoiesis. In contrast, recipient mice transplanted with a 5:5 ratio began to exhibit more myeloid-biased hematopoiesis eight weeks after transplantation, and by week 16, they produced significantly more myeloid cells than the other group (Fig. 6, B-D). Additionally, we conducted further investigations to determine whether lymphoid hematopoiesis could be accelerated by alteration of LT-HSC/ST-HSC ratios using cells from aged mice. To verify this, we isolated aged LT-HSCs and ST-HSCs and reconstituted them with a 5:5 ratio or a 2:8 ratio prior to transplantation (Fig. S6 A). Peripheral blood analysis revealed that lymphoid lineage output in the recipient mice transplanted with a 2:8 ratio was significantly greater than those with a 5:5 ratio (Fig. S6 B-D). Based on these findings, we concluded that myeloid-biased hematopoiesis observed following transplantation of aged HSCs was caused by a relative decrease in ST-HSC in the bulk-HSC compartment in aged mice rather than the selective expansion of myeloid-biased HSC clones.

Age-associated extramedullary changes accelerate myeloid-biased hematopoiesis

We found that the fluctuating LT-HSC/ST-HSC ratio in the bulk-HSC compartment corresponded with the myeloid-biased hematopoiesis associated with aging. In contrast, the degree of myeloid bias observed in mice older than 23 months without transplantation (Fig. 1 A; $49.0 \pm 23.4\%$) was significantly greater than in recipient mice receiving young mixed HSCs with a 5:5 ratio 16 weeks after transplantation (Fig. 6 D; $30.8 \pm 30.8\%$). This difference indicated that other intrinsic or extrinsic factors might exist in mice older than 23 months to promote myeloid-biased hematopoiesis. To investigate this further, we transplanted ten GFP⁺ young LT-HSCs into young or 2-year-old recipient mice and examined their PB (Fig. 7 A). We limited the observation period to 12 weeks because many aged recipient mice had died prior to that point (Fig. 7 B). PB analysis revealed that aged recipient mice receiving young LT-HSCs produced significantly more myeloid cells than young recipient mice (Fig. 7, C-E). These results suggested that recipient-dependent extrinsic factors, rather than intrinsic factors of transplanted HSCs, had a greater impact on hematopoietic cell differentiation. Then, we examined BM to determine what extrinsic factors affected differentiation of transplanted HSCs. We performed BM analysis for the mice denoted by † in Figure 7C because many of the aged mice had died before the analysis. Percentages of CMP and GMP, downstream components of myeloid differentiation, were stable or significantly lower in aged recipient mice than in young recipient mice, just as they were in non-treated aged mice (Fig. 7 F; Fig. 1 B). In addition, the percentage of CLP, a downstream component of lymphoid

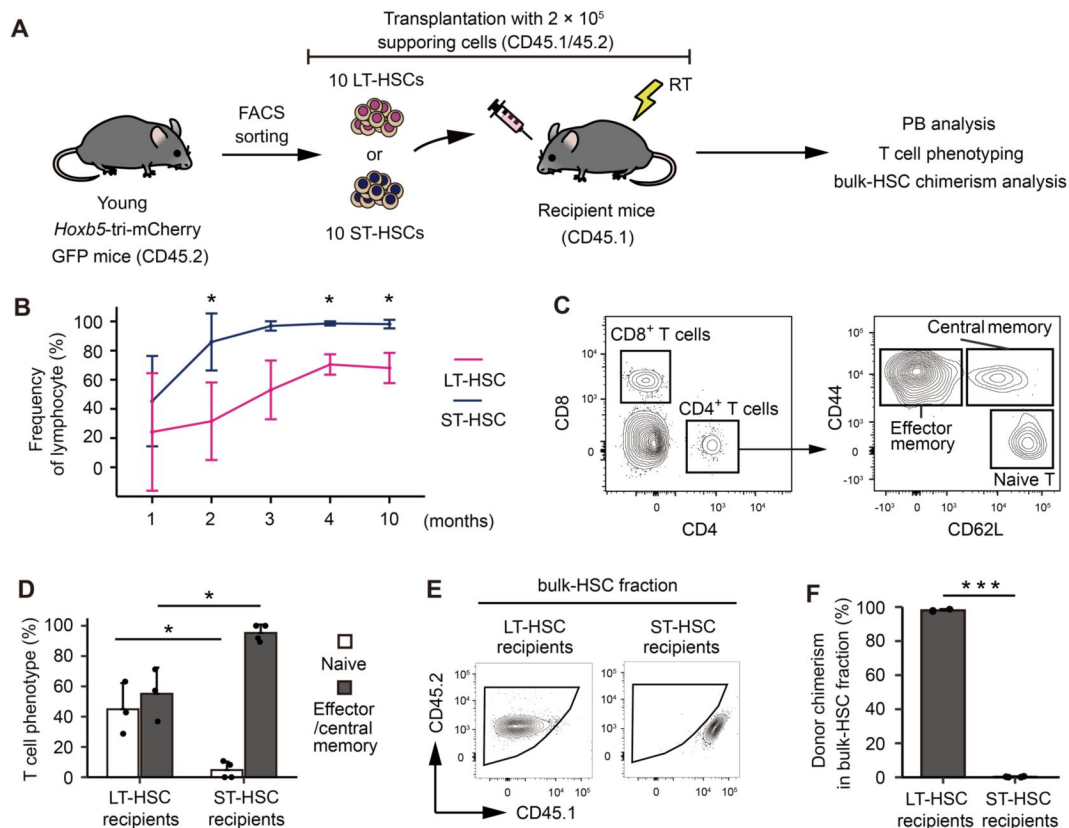


Figure 5.

The memory-type lymphocytes in the peripheral blood make it look as if ST-HSCs are lymphoid-biased HSCs.

(A) Experimental design for assessing the lineage output of young LT-HSCs or ST-HSCs.

Ten LT-HSCs, or ten ST-HSCs were isolated from 2-month-old CD45.2 *Hoxb5*-tri-mCherry GFP mice and were transplanted into lethally irradiated CD45.1 recipient mice with 2×10^5 supporting cells (LT-HSCs, $n = 3$; ST-HSCs, $n = 4$).

(B) Kinetics of average frequency of lymphoid cells (B cells, T cells, and NK cells) in donor fraction after LT-HSCs or ST-HSCs transplantation.

(C) Gating scheme to identify memory (Central and effector) T cells and naive T cells in the PB after excluding doublets, dead cells, and non-donor cells.

(D) Percentage of memory (Central and effector) T cells and naive T cells in donor CD4⁺ fraction 10 months after LT-HSC or ST-HSC transplantation.

(E) Gating scheme to identify donor cells in bulk-HSC fraction in bone marrow analysis.

(F) Donor chimerism in bulk-HSC fraction 12 months after LT-HSCs or ST-HSCs transplantation. $*P < 0.05$. $***P < 0.001$. Data and error bars represent means $\pm$ standard deviation.

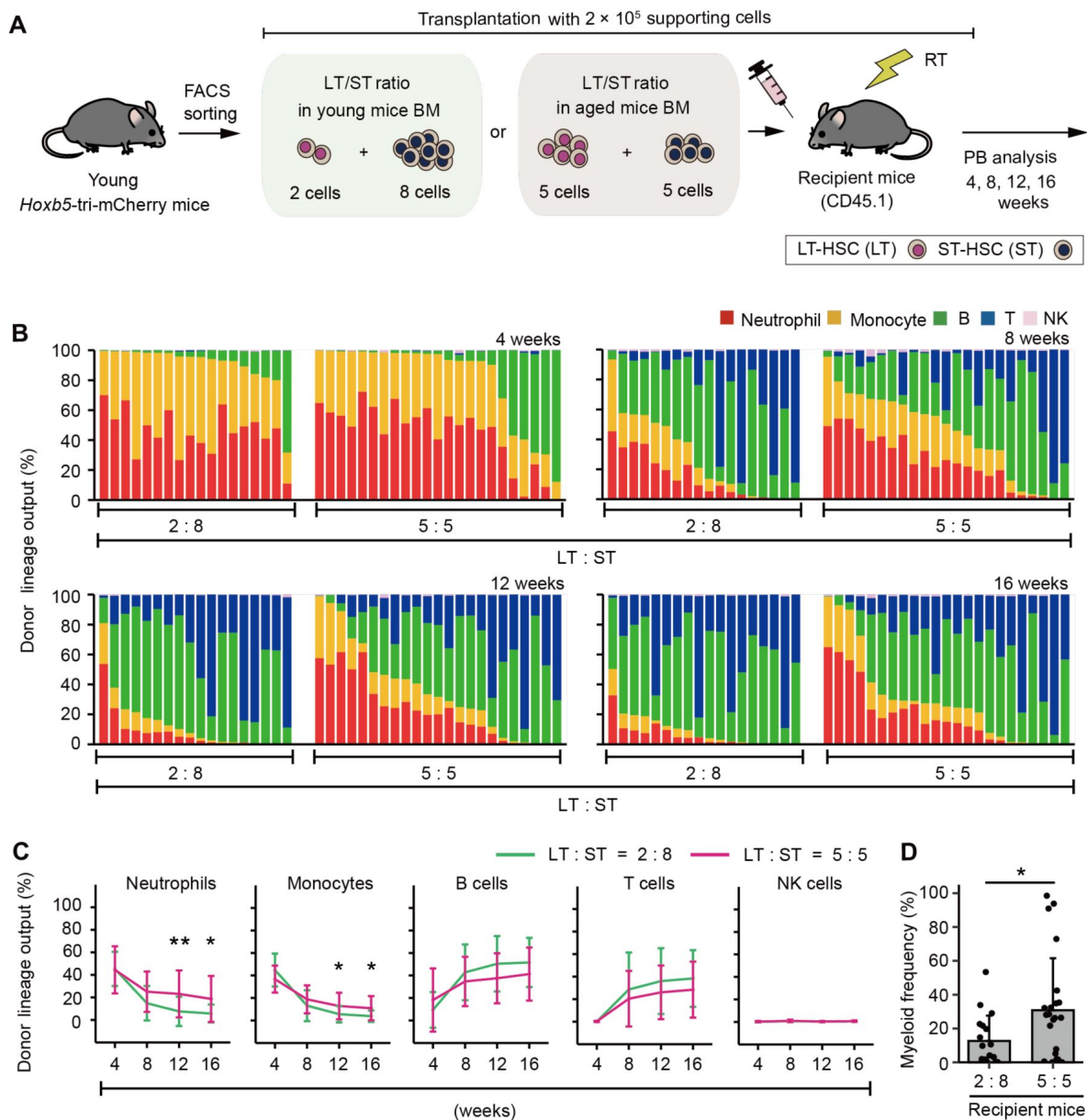


Figure 6.

Hematopoiesis after transplantation inclined either toward myeloid or lymphoid cell production by artificially changing the ratio of LT-HSC/ST-HSC.

(A) Experimental design for the transplantation of 2-3-month-old LT-HSCs and ST-HSCs in a 2:8 ratio (the same ratio as in young mice BM) or 5:5 ratio (the same ratio as in aged mice BM). Donor cells were transplanted with 2×10^5 CD45.1⁺/CD45.2⁺ supporting cells into lethally irradiated CD45.1⁺ recipient mice (2:8 ratio, $n = 18$; 5:5 ratio, $n = 23$).

(B) Donor lineage output of young LT-HSCs and ST-HSCs transplanted either in a 2:8 ratio or a 5:5 ratio at 4, 8, 12, 16 weeks post-transplant. Each bar represents an individual mouse.

(C) Kinetics of average lineage output of young LT-HSCs and ST-HSCs in a 2:8 ratio or a 5:5 ratio at 4, 8, 12, 16 weeks post-transplant.

(D) Frequency of myeloid cells in donor cell fraction. * $P < 0.05$. ** $P < 0.01$. Error bars represent standard deviation. Data represent two independent experiments.

differentiation, did not differ between young and aged recipients (**Fig. 7 F**). In contrast to previous studies (Pinho et al., 2018; Ergen et al., 2012), these findings suggested that intramedullary age-associated changes may not have a significant impact on LT-HSC differentiation.

Thymi and spleens of recipient mice were then examined, as these are the structures of lymphoid maturation and production of lymphoid progenitors following migration from BM and prior to their appearance in PB. The thymus and spleen analyses were also performed on the mice denoted by † in **Figure 7C**. Donor cells were not microscopically detectable in the majority of aged recipient mice (**Fig. 7, G and I**), and quantitative analysis using a flow cytometer revealed that the frequency of donor cells in spleens and thymi of aged recipients was significantly lower than in young recipients (**Fig. 7, H and J**). These results indicate that the process of lymphoid lineage differentiation is impaired in the spleens and thymi of aged mice compared to young mice, or that differentiating cells in the bone marrow do not successfully migrate into these secondary lymphoid organs. These factors contribute to the enhanced myeloid-biased hematopoiesis in peripheral blood due to a decrease in de novo lymphocyte production.

Discussion

Age-related, myeloid-biased hematopoiesis has been suggested as a potential reason for the decline in acquired immunity in the elderly (Geiger et al., 2013). This process has been linked to an increase in myeloid-biased HSC clones in the bulk-HSC fraction identified by surface antigens through aging. By isolating LT-HSCs and ST-HSCs, respectively, we discovered that age-related changes in the ratio of ST-HSCs to LT-HSCs in bulk-HSCs is responsible for myeloid-biased hematopoiesis.

When aged bulk-HSCs were transplanted into young mice, post-transplant hematopoiesis in recipient mice was significantly biased toward myeloid in PB compared to transplantation of young bulk-HSCs (Beerman et al., 2010; Dykstra et al., 2011; Sudo et al., 2000). Historically, this myeloid-biased hematopoiesis derived from aged bulk-HSCs has been considered as evidence supporting expansion of myeloid-biased HSC clones with aging (Geiger et al., 2013; de Haan and Lazare 2018). Other studies suggest that blockage of lymphoid hematopoiesis in aged mice results in myeloid-skewed hematopoiesis through alternative mechanisms. However, this result should be interpreted carefully, since Busulfan was used for myeloablative treatment in this study (Montecino-Rodriguez et al., 2019). To clarify the cause for this disparity, we chose to examine the LT-HSC fraction, which persists in the BM for long periods of time and is enriched for aging-related alterations. We discovered that post-transplant hematopoiesis of aged LT-HSCs did not vary from that of young LT-HSCs in terms of differentiation capacity. Expression of myeloid-related genes was not significantly altered when LT-HSCs from young and aged mice were compared. Therefore, we inferred that age-related myeloid-biased hematopoiesis cannot be attributed to an increase in myeloid-biased HSCs, at least among LT-HSCs. Given that bulk-HSCs consist of LT-HSCs and ST-HSCs, and that LT-HSCs exhibit no change in differentiation potential with age, we hypothesized that ST-HSCs may be responsible for the myeloid-skewing phenotype in PB. Transcriptomic analysis revealed no enrichment of myeloid-related genes in the ST-HSC fraction, ruling out the possibility of an increase in myeloid-biased HSCs in the ST-HSC fraction. Alternatively, we observed a proportionate decrease in ST-HSCs in bulk-HSCs accompanying an increase in LT-HSCs. We postulated that this relative decrease in ST-HSC ratio is responsible for myeloid-biased hematopoiesis. When isolated LT-HSCs and ST-HSCs from young mice were reconstituted to replicate an aged BM type in terms of the LT-HSC/ST-HSC ratio, we discovered that PB was strongly skewed towards myeloid cells after transplantation of reconstituted HSCs. In contrast, transplantation of reconstituted HSCs with a young BM type resulted in a significantly lymphoid-skewed PB profile.

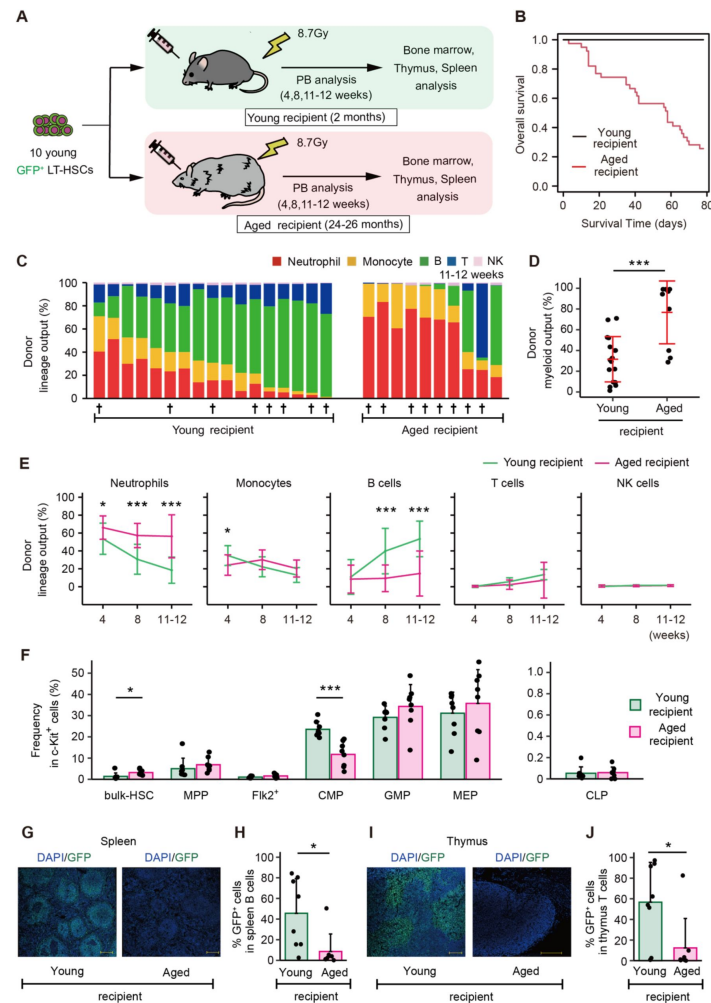


Figure 7.

Age-associated physiological changes drive differentiation of LT-HSCs toward myeloid cells.

(A) Experimental design for assessing the impact of age-associated physiological changes on differentiation of LT-HSCs. Ten GFP⁺ LT-HSCs sorted from young (2-3 months) *Hoxb5*-tri-mCherry GFP mice, were transplanted with 2×10^5 CD45.1⁺/CD45.2⁺ supporting cells into lethally irradiated young or aged recipient mice. We defined donor cells as GFP⁺ cells and supporting cells as CD45.1⁺/CD45.2⁺ cells.

(B) Survival rate of recipient mice in each group.

(C) Donor lineage output in young or aged recipient mice 11-12 weeks after transplanting young LT-HSCs (young recipient, $n = 17$; aged recipient, $n = 10$).

(D) Myeloid output (Frequency of donor myeloid cells in donor fraction) in young or aged recipient mice 11-12 weeks after transplantation.

(E) Kinetics of lineage output from donor LT-HSCs in young or aged recipient mice 4, 8, 11-12 weeks after transplantation.

(F) Average frequency of donor bulk-HSC and progenitor cells in donor c-Kit⁺ cells in BM (young recipient, $n = 8$; aged recipient, $n = 8$). BM samples were taken from mice denoted by † in [Figure 7C](#).

(G) Representative immunofluorescence images of frozen spleen sections derived from young or aged recipient mice. Green: donor cells (GFP fluorescence); blue: DNA (DAPI); Scale bar: 200 μ m.

(H) Frequency of donor cells in spleen B cells of young or aged recipient mice (young recipient, $n = 8$; aged recipient, $n = 8$). Spleens are taken from mice denoted by † in [Figure 7C](#).

(I) Representative immunofluorescence images of frozen thymus sections derived from young or aged recipient mice. Green: donor cells (GFP fluorescence); blue: DNA (DAPI); Scale bar: 200 μ m.

(J) Frequency of donor cells in thymus T cells of young or aged recipient mice (young recipient, $n = 8$; aged recipient, $n = 8$). Thymi are taken from mice denoted by † in [Figure 7C](#). * $P < 0.05$. *** $P < 0.001$. Error bars represent standard deviation. Data represent two independent experiments.

On the other hand, it has been reported that myeloid-biased HSCs have a more persistent capacity for self-renewal (Muller-Sieburg et al., 2004 [↗](#)). In fact, in our prior investigation comparing PB 16 weeks post-transplant, myeloid cell production in PB was greater when LT-HSCs were transplanted, while ST-HSC transplantation resulted in a considerable increase in lymphoid cell production (Chen et al., 2016 [↗](#)). It is therefore possible that the LT-HSCs and ST-HSCs we isolated utilizing the Hoxb5 reporter system are myeloid- and lymphoid-biased HSCs, respectively. However, the myeloid-producing capacity of LT-HSCs and ST-HSCs 4 weeks post-transplantation is equivalent (Chen et al., 2016 [↗](#)). In contrast, 16 weeks after transplantation, ST-HSC transplant recipients demonstrate negligible myeloid development and a predominance of lymphocytes, unlike LT-HSC transplant recipients (Chen et al., 2016 [↗](#)). These characteristics show that early post-transplant ST-HSCs lack the phenotypic of lymphoid-biased HSCs. When ST-HSC-transplanted BM cells are employed as donors for secondary transplantation, no donor-cell-derived hematopoiesis is detected in recipients (Chen et al., 2016 [↗](#)). In addition, HSCs are not observed in the BM following transplantation of ST-HSCs (Fig. 5 [↗](#)). These results indicate that ST-HSCs lose their capacity for self-renewal earlier after transplantation than LT-HSCs. In addition, phenotypic examination of lymphocytes in PB after ST-HSC transplantation revealed that nearly all of them were memory cells, and that there was no *de novo* supply of T cells. In other words, ST-HSCs may have been erroneously classified as lymphoid-biased HSCs due to the preservation of memory lymphocytes with a long half-life in the peripheral circulation, despite the absence of *de novo* hematopoiesis following transplantation of ST-HSCs. These findings suggest that the bias toward myeloid and lymphoid lineages in post-transplant PB is not regulated by heterogeneity of multipotency, but by heterogeneity of self-renewal capacity.

Multiple extrinsic factors have been implicated as HSC-independent causes of myeloid-biased hematopoiesis associated with aging. Some findings suggest that cell interaction with niche cells in BM is altered (Pinho et al., 2018 [↗](#)), whereas others claim that chronic inflammation in aged mice lowers production of lymphoid progenitors and induces myeloid-biased hematopoiesis (Montecino-Rodriguez et al., 2019 [↗](#); Ergen et al., 2012 [↗](#)). To identify HSC-independent causes of myeloid-biased hematopoiesis specifically, we transplanted young LT-HSCs into young or aged mice. Cellular differentiation observed in BM revealed that proportions of GMP, MEP, and CLP are unchanged between young and aged recipients. In contrast, an examination of spleens and thymi, the sites of lymphocyte maturation, revealed the absence of donor cells in aged recipients. These findings suggest that myeloid cells increase in aging mice due to a relative decrease in the ratio of mature lymphocytes in PB, resulting from inhibition of lymphocyte maturation outside the BM, such as in the thymus and spleen, rather than a significant change in cell differentiation caused by the intramedullary environment. Attempts have been made in the past to restore thymus function in aged mice by administering keratinocyte growth factor (KGF) (S. W. Rossi et al., 2007 [↗](#)). Indeed, in aged mice, lymphocyte hematopoiesis recovers, which is consistent with our results that the extramedullary environment influences the myeloid-bias phenotype in PB (S. W. Rossi et al., 2007 [↗](#)).

It has been clinically documented that recovery of the lymphocyte fraction in PB is delayed in elderly donors, compared to young donors (González-Vicent et al., 2017 [↗](#); Baron et al., 2006 [↗](#)). Although it is undeniable that this phenomenon may be due to decreased HSC hematopoietic potential, it may also be due to a change in the ratio of LT-HSC/ST-HSCs in bulk-HSCs, as described in our report, which may give the appearance of a myeloid bias in PB after transplantation. It has also been noted that lymphocyte recovery is delayed in elderly patients compared to younger patients (Berger et al., 2008 [↗](#); van der Maas et al., 2021 [↗](#)). Age-related loss in thymus and spleen functions has been recorded in humans, implying that the same mechanism is acting as in our mouse observations. If it is possible to assess and regulate the LT-HSC/ST-HSC ratio in donors and thymus/spleen function in recipients in the future, it will help to develop accurate models for predicting lymphocyte recovery after transplantation as well as new transplantation strategies.

In contrast, our findings should be considered in light of some limitations. In this report, we primarily performed ten to twenty cell transplantation assays. Therefore, the current theory should be revalidated using single-cell technology with lineage tracing system (Yamamoto et al. 2018 [↗](#); Rodriguez-Fraticelli et al. 2020 [↗](#)). This approach will investigate changes in the self-renewal capacity of individual HSCs and their subsequent differentiation into progenitor cells and peripheral blood cells. Additionally, in this report we purified LT-HSCs by Hoxb5 reporter system. In contrast, various LT-HSC markers have been previously reported. Therefore, it is ideal to validate our findings using other LT-HSC makers.

In summary, we have demonstrated for the first time that the ratio of LT-HSC/ST-HSC is important in age-related myeloid skewed hematopoiesis as an HSC-dependent factor and that aging-related thymus and spleen dysfunction also contributes significantly as an HSC-independent factor. Furthermore, here we propose a “self-renewal heterogeneity model” as a new mechanism for hematopoietic heterogeneity, including myeloid-biased hematopoiesis in aged mice (Fig. 8 [↗](#)). In addition to this report, we have also shown that cellular differentiation in the PB after transplantation can vary by changing transplantation conditions, ultimately altering self-renewal ability (Sakamaki et al., 2021 [↗](#); Nishi et al., 2022 [↗](#)). This highlights the significance of deciphering molecular processes that regulate the heterogeneity of HSC self-renewal capacity, which will help to provide a better understanding of the hematopoietic system and the hematopoietic hierarchy in the future.

Methods

Mice

Mice with an *Hoxb5*-tri-mCherry (C57BL/6J background), derived from our previous work (Chen et al., 2016 [↗](#)), were harvested for donor cells for transplantation, PB, and BM analysis. CAG-EGFP mice (C57BL/6J background) were bred with *Hoxb5*-tri-mCherry for transplantation experiments. 8-14-week-old C57BL/6-Ly5.1 mice (purchased from Sankyo Labo Service) were used as recipients for transplantation assays. For aged recipients, 2-year-old CD45.2 C57BL/6J mice were purchased from CLEA Japan. Supporting BM cells were collected from 8-12-week-old C57BL/6-Ly5.1 × C57BL/6J (F₁ mice CD45.1⁺/CD45.2⁺). All mice were housed in specific pathogen-free (SPF) conditions and were carefully observed by staff members. Mice were bred according to RIKEN or Kyoto University. All animal protocols were approved by the RIKEN Center for Biosystems Dynamics Research and Kyoto University.

Flow cytometry, cell sorting, and bone marrow analyses

Flow cytometry and cell sorting were performed on a FACS Aria II cell sorter (BD Biosciences) and analyzed using FlowJo™ software (BD Biosciences). Bone marrow (BM) cells were collected from bilateral tibias, femurs, humeri, and pelvises in Ca²⁺- and Mg²⁺-free PBS supplemented with 2% heat-inactivated bovine serum (Gibco) and 2 mM EDTA (Thermo Fisher Scientific). Cells were passed through 100-μm, and 40-μm strainers (Corning) before analysis and sorting. Prior to staining, samples were blocked with 50 ug/mL rat IgG (Sigma-Aldrich) for 15 min. To enrich HSCs and progenitor populations for sorting and bone marrow analyses, cells were stained with APC-eFluor780-conjugated anti-c-Kit (clone: 2B8) and fractionated using anti-APC magnetic beads and LS columns (both Miltenyi Biotec). c-Kit⁺ cells were then stained with combinations of antibodies as described in Supplemental Table 1. All staining was incubated at 4°C for 30 min, except for CD34 staining, which was incubated for 90 min. Samples were washed twice after staining. Prior to flow cytometry or cell sorting, samples were stained with SYTOX Red Dead Cell Stain (Life Technologies) or 7-aminoactinomycin D (BioLegend). Cells were double sorted for purity. For the combined analysis of the upstream (HSC, MPP, Flk2⁺) and downstream (CLP, MEP, CMP, GMP) fractions in Figures 1B [↗](#) and 7F [↗](#), we normalized by c-Kit⁺ cells because we performed a c-Kit enrichment for the bone marrow analysis.

Self-renewal heterogeneity model

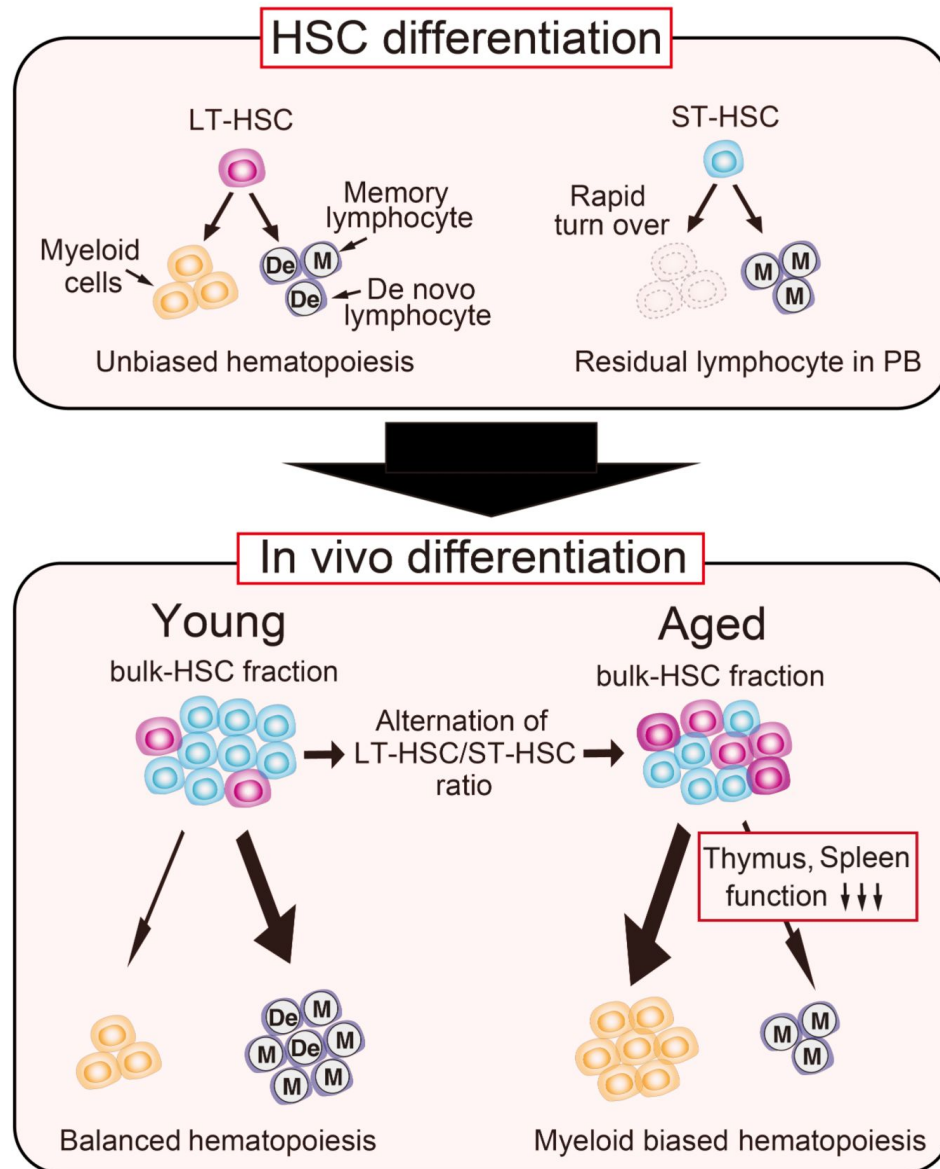


Figure 8.

Our new model: Self-renewal heterogeneity model.

It has been thought that there were myeloid (My-) or lymphoid biased (Ly-) HSCs, and that clonal selection of My-HSCs caused age-associated myeloid biased hematopoiesis. However, in our model, LT-HSCs represent unbiased hematopoiesis throughout life. ST-HSCs lose their hematopoietic ability within a short period and memory-type lymphocytes remains in the PB after ST-HSC transplantation. These remaining memory-type lymphocytes make it look as if ST-HSCs are lymphoid-biased (The upper section). As a result, the age-associated relative decrease of ST-HSCs in bulk-HSC fraction causes myeloid biased hematopoiesis with age. Additionally, the blockage of lymphoid differentiation at spleen and thymus accelerates further myeloid biased hematopoiesis in aged mice (The lower section).

Transplantation and peripheral blood analyses

12–24 hours prior to transplantation, C57BL/6-Ly5.1 mice, or aged C57BL/6J recipient mice were lethally irradiated with single doses of 8.7 Gy or 9.1 Gy. For transplantation assays, donor cells were first combined with 2×10^5 whole bone marrow supporting cells (C57BL/6-Ly5.1 $\times$ C57BL/6J F₁ mice CD45.1⁺/CD45.2⁺) in 200 μ L of PBS with 2% FBS, and then injected into the retro-orbital venous plexus. For evaluation of post-transplant kinetics, PB was collected and analyzed. At each time point, 50 μ L of blood were collected from the tail vein and re-suspended in Ca²⁺- and Mg²⁺-free PBS supplemented with 2 mM EDTA. Red blood cells were lysed twice on ice for 3 min with BD Pharm Lyse Buffer (BD Pharmingen). Identification of leukocyte subsets was performed by staining with antibodies. Antibody information was described in Supplemental Table 1. For evaluation of lineage output, the frequency of each lineage (NK cell, B cell, T cell, neutrophil, and monocyte) was determined in the whole fraction. The analysis of donor lineage output was restricted to donor cells showing $\geq 0.1\%$ donor chimerism at the last PB analysis to allow reliable detection. The percentage of donor chimerism in PB was defined as the percentage of CD45.1⁺CD45.2⁺ cells among total CD45.1⁺CD45.2⁺ and CD45.1⁺CD45.2⁺ cells. PB data represents mice which survived until the last PB analysis.

Thymus and Spleen analysis

Thymi and spleens were harvested and disrupted into a single cell suspension. Cells were passed through 100- μ m, and 40- μ m strainers (Corning). Isolated thymocytes and splenocytes were incubated on ice for 30 min with appropriately diluted antibodies in staining buffer. Identification of B cells in spleens and T cells in thymi was performed by staining with antibodies described in Supplemental Table 1. Prior to flow cytometry or cell sorting, samples were stained with SYTOX Red Dead Cell Stain (Life Technologies).

Tissue imaging

Freshly dissected spleens and thymi were fixed in 4% PFA (Nacalai tesque) in PBS for 24 h at 4°C. After PFA washout with PBS, tissues were cryoprotected with 30% sucrose in PBS for 24 h at 4°C, embedded in Tissue-Tek O.C.T. compound (Sakura Finetek), and snap-frozen in liquid nitrogen. Serial 10- μ m longitudinal cryostat sections were obtained using CryoStar NX50 (Thermo Fisher Scientific). Cell nuclei were counterstained with 4',6-diamidino-2-phenylindole (DAPI, 1 μ g/mL, Roche). To reduce autofluorescence, tissue sections were treated with Vector TrueVIEW autofluorescence quenching kit (Vector Laboratories). Images were obtained from Leica TCS SP8 (Leica).

RNA sequencing

Total RNA was isolated with Trizol (Thermo Fisher Scientific) and cleaned up using RNeasy MinElute columns (QIAGEN). cDNA libraries were prepared from bulk-HSCs, ST-HSCs and LT-HSCs using a KAPA RNA HyperPrep kit with RiboErase (HMR) (Kapa Biosystems) and sequenced using a HiSeq 1500 (Illumina) to obtain 2×127 base-pair (bp) paired-end reads. Three replicates were sequenced for each population. Raw transcriptome sequence data were mapped to the genome (mm10) using HISAT2 (ver 2.1.0) (Kim et al., 2015 [DOI](#)). Alignments were then passed to StringTie (ver 1.3.4d), which was used to assemble and quantify transcripts in each sample (Pertea et al., 2016 [DOI](#)). EdgeR (ver 3.22.5) was then used to compare all transcripts across conditions and to produce tables and plots of differentially expressed genes and transcripts (Robinson et al., 2009 [DOI](#)). Normalization with the TMM method was performed with the edgeR package in Bioconductor (<https://bioconductor.org/> [DOI](#)). Genes (12,808 genes) were selected for hierarchical clustering based on the mean of TMM normalized read counts across all samples' cells (mean ≥ 1). Then, clustering was performed by using the hclust function for R (distance = Spearman's correlation; method = 'ward.D2'). We depicted Venn diagrams and performed GSEA analyses in

Figures 4D-G and S4C-D using previously published gene sets after excluding genes that could not be annotated by our transcriptome dataset. GSEA was performed using GSEA software (<http://www.broadinstitute.org/gsea>) with default settings (Subramanian et al., 2005).

Quantification and statistical analyses

Statistical analyses were performed using ggplot2 in R (version 4.1.2) or Microsoft Excel. Sample size for each experiment and replicate numbers of experiments are included in figure legends. Statistical significance was determined using Welch's t test. *P* values < 0.05 were considered significant.

Data and materials availability

Sequencing data has been deposited in the Gene Expression Omnibus under accession code GSE226803. Correspondence and requests for materials should be addressed to M.M. (miya75@med.kobe-u.ac.jp).

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Additional information

Authorship Contributions

K.N. conceived, performed, analyzed the experiments. T.S. and K.S. performed transplantation experiments. M.M. conceived, performed, analyzed, and oversaw the experiments. K.N. and M.M. wrote the manuscript. T.S., A.N., K.S.K., K.S., M.A., N.Y., and A.T. provided comments on the manuscript.

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Author information

Katsuyuki Nishi

Hematopoietic Stem Cell Biology and Medical Innovation (HSCBMD), Department of Pediatrics, Kobe University Graduate School of Medicine, Kobe, Japan, RIKEN Center for Biosystems Dynamics Research, Kobe, Japan

Taro Sakamaki

Hematopoietic Stem Cell Biology and Medical Innovation (HSCBMD), Department of Pediatrics, Kobe University Graduate School of Medicine, Kobe, Japan, RIKEN Center for Biosystems Dynamics Research, Kobe, Japan

Akiomi Nagasaka

Hematopoietic Stem Cell Biology and Medical Innovation (HSCBMD), Department of Pediatrics, Kobe University Graduate School of Medicine, Kobe, Japan, RIKEN Center for Biosystems Dynamics Research, Kobe, Japan

Kevin S Kao

Weill Cornell, Rockefeller, Sloan-Kettering, Tri-Institutional MD-PhD Program, New York, United States

Kay Sadaoka

Hematopoietic Stem Cell Biology and Medical Innovation (HSCBMD), Department of Pediatrics, Kobe University Graduate School of Medicine, Kobe, Japan, RIKEN Center for Biosystems Dynamics Research, Kobe, Japan

Masahide Asano

Institute of Laboratory Animals, Kyoto University Graduate School of Medicine, Kyoto, Japan

Nobuyuki Yamamoto

Department of Pediatrics, Kobe University Graduate School of Medicine, Kobe, Japan
ORCID iD: [0000-0002-1006-7394](https://orcid.org/0000-0002-1006-7394)

Akifumi Takaori-Kondo

Department of Hematology and Oncology, Kyoto University Graduate School of Medicine, Kyoto, Japan
ORCID iD: [0000-0001-7678-4284](https://orcid.org/0000-0001-7678-4284)

Masanori Miyanishi

Hematopoietic Stem Cell Biology and Medical Innovation (HSCBMD), Department of Pediatrics, Kobe University Graduate School of Medicine, Kobe, Japan, RIKEN Center for Biosystems Dynamics Research, Kobe, Japan

For correspondence: miya75@med.kobe-u.ac.jp

Editors

Reviewing Editor

Jian Xu

St. Jude Children's Research Hospital, Memphis, United States of America

Reviewer #1 (Public review):

In this study, Nishi et al. claim that the ratio of long-term hematopoietic stem cell (LT-HSC) versus short-term HSC (ST-HSC) determines the lineage output of HSCs and reduced ratio of ST-HSC in aged mice causes myeloid-biased hematopoiesis. Authors used Hoxb5 reporter mice to isolated LT-HSC and ST-HSC and performed molecular analyses and transplantation assays to support their arguments. How hematopoietic system becomes myeloid-biased upon aging is an important question with many implications in disease context as well. However, this study needs more definitive data.

(1) Authors' experimental designs have some caveats to definitely support their claims. Authors claimed that aged LT-HSCs have no myeloid-biased clone expansion using transplantation assays. In these experiments, authors used 10 HSCs and young mice as recipients. Given the huge expansion of old HSC by number and known heterogeneity in immunophenotypically defined HSC populations, it is questionable how 10 out of so many old HSCs (an average of 300,000 up to 500,000 cells per mouse; Mitchell et al., Nature Cell Biology, 2023) can faithfully represent old HSC population. The Hoxb5+ old HSC primary and secondary recipient mice data (Fig. 2C and D) support this concern. In addition, they only used young recipients. Considering the importance of inflammatory aged niche in the myeloid-biased lineage output, transplanting young vs old LT-HSCs into aged mice will complete the whole picture.

In response to the above comments, the authors calculated the required sample size as approximately 384 cells to represent 500,000 HSCs per old mouse. Based on the total 1260 cells used throughout the whole manuscript (Figures 2, 3, 5, 6, S3, and S6), the authors claimed that the data is reflecting old HSC behavior. However, 384 cells represent HSCs from one old mouse. Following the authors' logic, they did only 3.2 mice (1260/384) experiment for the whole manuscript to make their argument. N of 3 is not enough, especially for old mice experiments considering the heterogeneity of aged mice. Also, they did not address the comment regarding inflammatory aged niche effects.

(2) Authors' molecular data analyses need more rigor with unbiased approaches. They claimed that neither aged LT-HSCs nor aged ST-HSCs exhibited myeloid or lymphoid gene set enrichment but aged bulk HSCs, which are just a sum of LT-HSCs and ST-HSCs by their gating scheme (Fig. 4A), showed the "tendency" of enrichment of myeloid-related genes based on the selected gene set (Fig. 4D). Although the proportion of ST-HSCs is reduced in bulk HSCs upon aging, since ST-HSCs do not exhibit lymphoid gene set enrichment based on their data, it is hard to understand how aged bulk HSCs have more myeloid gene set enrichment compared to young bulk HSCs. This bulk HSC data rather suggest that there could be a trend toward certain lineage bias (although not significant) in aged LT-HSCs or ST-HSCs. Authors need to verify the molecular lineage priming of LT-HSCs and ST-HSCs using another comprehensive dataset.

(3) Although authors could not find any molecular evidence for myeloid-biased hematopoiesis from old HSCs (either LT or ST), they argued that the ratio between LT-HSC and ST-HSC causes myeloid-biased hematopoiesis upon aging based on young HSC experiments (Fig. 6). However, old ST-HSC functional data showed that they barely contribute to blood production unlike young Hoxb5- HSCs (ST-HSC) in the transplantation setting (Fig. 2). Is there any evidence that in unperturbed native old hematopoiesis, old Hoxb5- HSCs (ST-HSC) still contribute to blood production? To answer this question, authors performed additional experiments with increased cell number (Fig. S6). Although Fig. S6.D data has a statistical significance, it is questionable how biologically meaningful it is. More fundamental question

is back to the representability. Can this cell number used in this experiment represent old HSC (either LT or ST) behavior?

<https://doi.org/10.7554/eLife.95880.3.sa3>

Reviewer #2 (Public review):

Summary:

Nishi et al, investigate the well-known and previously described phenomenon of age-associated myeloid-biased hematopoiesis. Using a previously established HoxB5mCherry mouse model, they used HoxB5⁺ and HoxB5⁻ HSCs to discriminate cells with long-term (LT-HSCs) and short-term (ST-HSCs) reconstitution potential and compared these populations to immunophenotypically defined 'bulk HSCs' that consists of a mixture of LT-HSC and ST-HSCs. They then isolated these HSC populations from young and aged mice to test their function and myeloid bias in non-competitive and competitive transplants into young and aged recipients. Based on quantification of hematopoietic cell frequencies in the bone marrow, peripheral blood, and in some experiments the spleen and thymus, the authors argue against the currently held belief that myeloid-biased HSCs expand with age.

While aspects of their work are fascinating and might have merit, several issues weaken the overall strength of the arguments and interpretation. Multiple experiments were done with a very low number of recipient mice, showed very large standard deviations, and had no statistically detectable difference between experimental groups. While the authors conclude that these experimental groups are not different, the displayed results seem too variable to conclude anything with certainty. The sensitivity of the performed experiments (e.g. Fig 3; Fig 6C, D) is too low to detect even reasonably strong differences between experimental groups and is thus inadequate to support the author's claims. This weakness of the study is not acknowledged in the text and is also not discussed. To support their conclusions the authors need to provide higher n-numbers and provide a detailed power analysis of the transplants in the methods section.

As the authors attempt to challenge the current model of the age-associated expansion of myeloid-biased HSCs (which has been observed and reproduced by many different groups), ideally additional strong evidence in the form of single-cell transplants is provided. It is also unclear why the authors believe that the observed reduction of ST-HSCs relative to LT-HSCs explains the myeloid-biased phenotype observed in the peripheral blood. This point seems counterintuitive and requires further explanation.

Based on my understanding of the presented data, the authors argue that myeloid-biased HSCs do not exist, as:

- a) they detect no difference between young/aged HSCs after transplant (mind low n-numbers and large std);
- b) myeloid progenitors downstream of HSCs only show minor or no changes in frequency and
- c) aged LT-HSCs do not outperform young LT-HSC in myeloid output LT-HSCs in competitive transplants (mind low n-numbers and large std!!!).

However, given the low n-numbers and high variance of the results, the argument seems weak and the presented data does not support the claims sufficiently. That the number of downstream progenitors does not change could be explained by other mechanisms, for instance, the frequently reported differentiation short-cuts of HSCs and/or changes in the microenvironment.

Strengths:

The authors present an interesting observation and offer an alternative explanation of the origins of aged-associated myeloid-biased hematopoiesis. Their data regarding the role of the microenvironment in the spleen and thymus appears to be convincing.

Weaknesses:

"Then, we found that the myeloid lineage proportions from young and aged LT-HSCs were nearly comparable during the observation period after transplantation (Fig. 3, B and C)." [Comment to the authors]: Given the large standard deviation and low n-numbers, the power of the analysis to detect differences between experimental groups is very low. Experimental groups with too large standard deviations (as displayed here) are difficult to interpret and might be inconclusive. The absence of clearly detectable differences between young and aged transplanted HSCs could thus simply be a false-negative result. The shown experimental results hence do not provide strong evidence for the author's interpretation of the data. The authors should add additional transplants and include a detailed power analysis to be able to detect differences between experimental groups with reasonable sensitivity.

Line 293: "Based on these findings, we concluded that myeloid-biased hematopoiesis observed following transplantation of aged HSCs was caused by a relative decrease in ST-HSC in the bulk-HSC compartment in aged mice rather than the selective expansion of myeloid-biased HSC clones."

[Comment to the authors]: Couldn't that also be explained by an increase in myeloid-biased HSCs, as repeatedly reported and seen in the expansion of CD150⁺ HSCs? It is not intuitively clear why a reduction of ST-HSCs clones would lead to a myeloid bias. The author should try to explain more clearly where they believe the increased number of myeloid cells comes from. What is the source of myeloid cells if the authors believe they are not derived from the expanded population of myeloid-biased HSCs?

New comment for the authors:

While the authors provide new evidence, clarify the text, and adjust their interpretation, the presented data remain weak and do not convincingly challenge the current paradigm. As myeloid-biased HSC expansion with age has been observed and published by many different groups, the authors need to provide much stronger evidence to challenge the observations of others. Key experiments that might support their claims had been suggested, but as indicated, the authors plan to provide these much more rigorous experiments in future studies. As it stands, the overall conclusions of this manuscript thus remain weak and preliminary.

In an attempt to quantify the absolute cell number of HSPC subpopulations, the authors use a usual readout and quantify "Number of cells per minute of analysis time". This appears to be a quick and dirty reanalysis of already existing flow cytometry data. Unfortunately, this quantification cannot count the absolute number of cells reliably, as the number of cells per minute recorded is heavily influenced by the abundance of other cell populations. Instead, the author should have counted the absolute number of HSCs, MPPs, GMPs, etc. per femur, which is typically done to address this question.

At this point, as authors are seemingly not willing to provide additional hard evidence to support their claims in this study and are instead in the process of preparing additional data for a future manuscript, I believe this study, as it stands (although weak), suggests an interesting alternative model. Despite being highly controversial, this alternative model warrants future investigations and discussions in the field. As always, it will also be important to reproduce these findings independently in other labs. As my concerns and the concerns of the other reviewers are documented and available to read by others, I believe the manuscript should be published in its current form to stimulate critical discussion and future investigations of the current model.

Reviewer #3 (Public review):

In this manuscript, Nishi et al. propose a new model to explain the previously reported myeloid-biased hematopoiesis associated with aging. Traditionally, this phenotype has been explained by the expansion of myeloid-biased hematopoietic stem cell (HSC) clones during aging. Here, the authors question this idea and show how their Hoxb5 reporter model can discriminate long-term (LT) and short-term (ST) HSC and characterized their lineage output after transplant. From these analyses, the authors conclude that changes during aging in the LT/ST HSC proportion explain the myeloid bias observed.

Comments on revisions:

I appreciate the authors' reply to some of my comments. However, there are some key aspects that remain unresolved. Please see below.

- The authors propose a critical change in the way we consider the mechanisms leading to lineage biased hematopoiesis during aging. As Reviewer 2 mentioned, such a strong claim needs to be supported by solid experimental data. Unfortunately, the level of variability in key in vivo experiments (Figure 2 and 3) diminishes the robustness of these results.

The authors argue that even with the low number of mice used in some of these experiments and the high level of variability, differences still reach (or not) statistical significance according to their analysis. I am not an expert on statistics but the only test that is mentioned is their methodology is a Welch's t test, which is only appropriate for data following a normal distribution. A more rigorous statistical analysis should be performed to sustain the claims included in the current manuscript.

- The chosen irradiation regiment might contribute to the uncertainty of the data and influence their interpretation. As the authors show in their response to my "comment to our #3-4 response", there is a considerable (and variable) amount of "radioresistant" CD45.1+CD45.2- cells in their primary recipients, which become concerningly high in the secondary transplant. This is not found in previous publications focused on this topic and, therefore, it makes it difficult to compare those studies with the present manuscript. The inclusion of this aspect in the text is appreciated but definitely reduces the impact of their claims.

- The correction introduced in the main text as an answer to the original comment #3-6 is still misleading. There is an assumption for GMP, CMP and MEP to increase with age if myeloid-biased HSC clones increase with age ("in contrast to what we anticipated"). Again, the link between these two changes could be more complex than just a direct correlation.

<https://doi.org/10.7554/eLife.95880.3.sa1>

Author response:

The following is the authors' response to the previous reviews

Public Reviews:

Reviewer #1 (Public review):

(1) Authors' experimental designs have some caveats to definitely support their claims. Authors claimed that aged LT-HSCs have no myeloid-biased clone expansion using

transplantation assays. In these experiments, authors used 10 HSCs and young mice as recipients. Given the huge expansion of old HSC by number and known heterogeneity in immunophenotypically defined HSC populations, it is questionable how 10 out of so many old HSCs (an average of 300,000 up to 500,000 cells per mouse; Mitchell et al., Nature Cell Biology, 2023) can faithfully represent old HSC population. The Hoxb5+ old HSC primary and secondary recipient mice data (Fig. 2C and D) support this concern. In addition, they only used young recipients. Considering the importance of inflammatory aged niche in the myeloid-biased lineage output, transplanting young vs old LT-HSCs into aged mice will complete the whole picture.

We sincerely appreciate your insightful comment regarding the existence of approximately 500,000 HSCs per mouse in older mice. To address this, we have conducted a statistical analysis to determine the appropriate sample size needed to estimate the characteristics of a population of 500,000 cells with a 95% confidence level and a $\pm 5\%$ margin of error. This calculation was performed using the finite population correction applied to Cochran's formula.

For our calculations, we used a proportion of 50% ($p = 0.5$), as it has been reported that approximately 50% of HSCs are myeloid-biased^{1,2}. The formula used is as follows:

$$n = \frac{N \cdot Z^2 \cdot p \cdot (1-p)}{(N-1) \cdot e^2 + Z^2 \cdot p \cdot (1-p)}$$

$N = 500,000$ (total population size)

$Z = 1.96$ (Z-score for a 95% confidence level)

$p = 0.5$ (expected proportion)

$e = 0.05$ (margin of error)

Applying this formula, we determined that the required sample size is approximately 384 cells. This sample size ensures that the observed proportion in the sample will reflect the characteristics of the entire population. In our study, we have conducted functional experiments across Figures 2, 3, 5, 6, S3, and S6, with a total sample size of $n = 126$, which corresponds to over 1260 cells. While it would be ideal to analyze all 500,000 cells, this would necessitate the use of 50,000 recipient mice, which is not feasible. We believe that the number of cells analyzed is reasonable from a statistical standpoint.

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- (2) Beerman, Isabel et al. "Functionally distinct hematopoietic stem cells modulate hematopoietic lineage potential during aging by a mechanism of clonal expansion." *Proceedings of the National Academy of Sciences of the United States of America* vol. 107,12 (2010): 5465-70. doi:10.1073/pnas.1000834107

(2) Authors' molecular data analyses need more rigor with unbiased approaches. They claimed that neither aged LT-HSCs nor aged ST-HSCs exhibited myeloid or lymphoid gene set enrichment but aged bulk HSCs, which are just a sum of LTHSCs and ST-HSCs by their gating scheme (Fig. 4A), showed the "tendency" of enrichment of myeloid-related genes based on the selected gene set (Fig. 4D). Although the proportion of ST-HSCs is reduced in bulk HSCs upon aging, since STHSCs do not exhibit lymphoid gene set enrichment based

on their data, it is hard to understand how aged bulk HSCs have more myeloid gene set enrichment compared to young bulk HSCs. This bulk HSC data rather suggest that there could be a trend toward certain lineage bias (although not significant) in aged LT-HSCs or ST-HSCs. Authors need to verify the molecular lineage priming of LT-HSCs and ST-HSCs using another comprehensive dataset.

Thank you for your thoughtful feedback regarding the lack of myeloid or lymphoid gene set enrichment in aged LT-HSCs and aged ST-HSCs, despite the observed tendency for myeloid-related gene enrichment in aged bulk HSCs.

First, we acknowledge that the GSEA results vary among the different myeloid gene sets analyzed (Fig. 4, D–F; Fig. S4, C–D). Additionally, a comprehensive analysis of mouse HSC aging using multiple RNA-seq datasets reported that nearly 80% of differentially expressed genes show poor reproducibility across datasets[1]. These factors highlight the challenges of interpreting lineage bias in HSCs based solely on previously published transcriptomic data.

Given these points, we believe that emphasizing functional experimental results is more critical than incorporating an additional dataset to support our claim. In this regard, we have confirmed that young and aged LT-HSCs have similar differentiation capacity (Figure 3), while myeloid-biased hematopoiesis is observed in aged bulk HSCs (Figure S3). These findings are further corroborated by independent functional experiments. We sincerely appreciate your insightful comments.

Reference

(1) Flohr Svendsen, Arthur et al. “A comprehensive transcriptome signature of murine hematopoietic stem cell aging.” *Blood* vol. 138,6 (2021): 439-451. doi:10.1182/blood.2020009729

(3) Although authors could not find any molecular evidence for myeloid-biased hematopoiesis from old HSCs (either LT or ST), they argued that the ratio between LT-HSC and ST-HSC causes myeloid-biased hematopoiesis upon aging based on young HSC experiments (Fig. 6). However, old ST-HSC functional data showed that they barely contribute to blood production unlike young Hoxb5- HSCs (ST-HSC) in the transplantation setting (Fig. 2). Is there any evidence that in unperturbed native old hematopoiesis, old Hoxb5- HSCs (ST-HSC) still contribute to blood production?

If so, what are their lineage potential/output? Without this information, it is hard to argue that the different ratio causes myeloid-biased hematopoiesis in aging context.

Thank you for the insightful and important question. The post-transplant chimerism of ST-HSCs was low in Fig. 2, indicating that transplantation induced a short-term loss of hematopoietic potential due to hematopoietic stress per cell.

To reduce this stress, we increased the number of HSCs in transplantation setting. In Fig. S6, old LT-HSCs and old ST-HSCs were transplanted in a 50:50 or 20:80 ratio, respectively. As shown in Fig. S6.D, the 20:80 group, which had a higher proportion of old ST-HSCs, exhibited a statistically significant increase in the lymphoid percentage in the peripheral blood post-transplantation.

These findings suggest that old ST-HSCs contribute to blood production following transplantation.

Reviewer #2 (Public review):

While aspects of their work are fascinating and might have merit, several issues weaken the overall strength of the arguments and interpretation. Multiple experiments were done with a very low number of recipient mice, showed very large standard deviations,

and had no statistically detectable difference between experimental groups. While the authors conclude that these experimental groups are not different, the displayed results seem too variable to conclude anything with certainty. The sensitivity of the performed experiments (e.g. Fig 3; Fig 6C, D) is too low to detect even reasonably strong differences between experimental groups and is thus inadequate to support the author's claims. This weakness of the study is not acknowledged in the text and is also not discussed. To support their conclusions the authors need to provide higher n-numbers and provide a detailed power analysis of the transplants in the methods section.

Response #2-1:

Thank you for your important remarks. The power analysis for this experiment shows that power = 0.319, suggesting that more number may be needed. On the other hand, our method for determining the sample size in Figure 3 is as follows:

- (1) First, we checked whether myeloid biased change is detected in the bulk-HSC fraction (Figure S3). The results showed that the difference in myeloid output at 16 weeks after transplantation was statistically significant (young vs. aged = 7.2 ± 8.9 vs. $42.1 \pm 35.5\%$, $p = 0.01$), even though $n = 10$.
- (2) Next, myeloid biased HSCs have been reported to be a fraction with high selfrenewal ability (2004, Blood). If myeloid biased HSCs increase with aging, the increase in myeloid biased HSCs in LT-HSC fraction would be detected with higher sensitivity than in the bulk-HSC fraction used in Figure S3.
- (3) However, there was no difference not only in p-values but also in the mean itself, young vs aged = $51.4 \pm 31.5\%$ vs $47.4 \pm 39.0\%$, $p = 0.82$, even though $n = 8$ in Figure 3. Since there was no difference in the mean itself, it is highly likely that no difference will be detected even if n is further increased.

Regarding Figure 6, we obtained a statistically significant difference and consider the sample size to be sufficient. In addition, we have performed various functional experiments (Figures 2, 5, 6 and S6), and have obtained consistent results that expansion of myeloid biased HSCs does not occur with aging in Hoxb5+HSCs fraction. Based on the above, we conclude that the LT-HSC fraction does not differ in myeloid differentiation potential with aging.

As the authors attempt to challenge the current model of the age-associated expansion of myeloid-biased HSCs (which has been observed and reproduced by many different groups), ideally additional strong evidence in the form of single-cell transplants is provided.

Response #2-2:

Thank you for the comments. As the reviewer pointed out, we hope we could reconfirm our results using single-cell level technology in the future.

On the other hand, we have reported that the ratio of myeloid to lymphoid cells in the peripheral blood changes when the number of HSCs transplanted, or the number of supporting cells transplanted with HSCs, is varied[1-2]. Therefore, single-cell transplant data need to be interpreted very carefully to determine differentiation potential.

From this viewpoint, future experiments will combine the Hoxb5 reporter system with a lineage tracing system that can track HSCs at the single-cell level over time. This approach will investigate changes in the self-renewal capacity of individual HSCs and their subsequent differentiation into progenitor cells and peripheral blood cells. We have reflected this comment by adding the following sentences in the manuscript.

[P19, L451] “In contrast, our findings should be considered in light of some limitations. In this report, we primarily performed ten to twenty cell transplantation assays. Therefore, the current theory should be revalidated using single-cell technology with lineage tracing system[3-4]. This approach will investigate changes in the self-renewal capacity of individual HSCs and their subsequent differentiation into progenitor cells and peripheral blood cells.”

It is also unclear why the authors believe that the observed reduction of ST-HSCs relative to LT-HSCs explains the myeloid-biased phenotype observed in the peripheral blood. This point seems counterintuitive and requires further explanation.

Response #2-3:

Thank you for your comment. We apologize for the insufficient explanation. Our data, as shown in Figures 3 and 4, demonstrate that the differentiation potential of LT-HSCs remains unchanged with age. Therefore, rather than suggesting that an increase in LT-HSCs with a consistent differentiation capacity leads to myeloid-biased hematopoiesis, it seems more accurate to highlight that the relative decrease in the proportion of ST-HSCs, which remain in peripheral blood as lymphocytes, leads to a relative increase in myeloid cells in peripheral blood and thus causes myeloid-biased hematopoiesis.

However, if we focus on the increase in the ratio of LT-HSCs, it is also plausible to explain that “with aging, the proportion of LT-HSCs capable of long-term myeloid hematopoiesis increases. As a result, from 16 weeks after transplantation, the influence of LT-HSCs maintaining the long-term ability to produce myeloid cells becomes relatively more significant, leading to an increase in the ratio of myeloid cells in the peripheral blood and causing myeloid-biased hematopoiesis.”

Based on my understanding of the presented data, the authors argue that myeloid-biased HSCs do not exist, as

a) they detect no difference between young/aged HSCs after transplant (mind low n-numbers and large std!!!); b) myeloid progenitors downstream of HSCs only show minor or no changes in frequency and c) aged LT-HSCs do not outperform young LT-HSC in myeloid output LT-HSCs in competitive transplants (mind low n-numbers and large std!!!).

However, given the low n-numbers and high variance of the results, the argument seems weak and the presented data does not support the claims sufficiently. That the number of downstream progenitors does not change could be explained by other mechanisms, for instance, the frequently reported differentiation short-cuts of HSCs and/or changes in the microenvironment.

Response #2-4:

We appreciate the comments. As mentioned above, we will correct the manuscript regarding the sample size. Regarding the interpreting of the lack of increase in the percentage of myeloid progenitor cells in the bone marrow with age, it is instead possible that various confounding factors, such as differentiation shortcuts or changes in the microenvironment, are involved.

However, even when aged LT-HSCs and young LT-HSCs are transplanted into the same recipient mice, the timing of the appearance of different cell fractions in peripheral blood is similar (Figure 3 of this paper). Therefore, we have not obtained data suggesting that clear shortcuts exist in the differentiation process of aged HSCs into neutrophils or monocytes. Additionally, it is currently consensually accepted that myeloid cells, including neutrophils and monocytes, differentiate from GMPs[1]. Since there is no changes in the proportion of

GMPs in the bone marrow with age, we concluded that the differentiation potential into myeloid cells remains consistent with aging.

"Then, we found that the myeloid lineage proportions from young and aged LT-HSCs were nearly comparable during the observation period after transplantation (Fig. 3, B and C)."

[Comment to the authors]: Given the large standard deviation and low n-numbers, the power of the analysis to detect differences between experimental groups is very low. Experimental groups with too large standard deviations (as displayed here) are difficult to interpret and might be inconclusive. The absence of clearly detectable differences between young and aged transplanted HSCs could thus simply be a false-negative result. The shown experimental results hence do not provide strong evidence for the author's interpretation of the data. The authors should add additional transplants and include a detailed power analysis to be able to detect differences between experimental groups with reasonable sensitivity.

Response #2-5:

Thank you for providing these insights. Regarding the sample size, we have addressed this in Response #2-1.

Line 293: "Based on these findings, we concluded that myeloid-biased hematopoiesis observed following transplantation of aged HSCs was caused by a relative decrease in ST-HSC in the bulk-HSC compartment in aged mice rather than the selective expansion of myeloid-biased HSC clones."

Couldn't that also be explained by an increase in myeloid-biased HSCs, as repeatedly reported and seen in the expansion of CD150+ HSCs? It is not intuitively clear why a reduction of ST-HSCs clones would lead to a myeloid bias. The author should try to explain more clearly where they believe the increased number of myeloid cells comes from. What is the source of myeloid cells if the authors believe they are not derived from the expanded population of myeloid-biased HSCs? t

Response #2-6:

Thank you for pointing this out. We apologize for the insufficient explanation. We will explain using Figure 8 from the paper.

First, our data show that LT-HSCs maintain their differentiation capacity with age, while ST-HSCs lose their self-renewal capacity earlier, so that only long-lived memory lymphocytes remain in the peripheral blood after the loss of selfrenewal capacity in ST-HSCs (Figure 8, upper panel). In mouse bone marrow, the proportion of LT-HSCs increases with age, while the proportion of ST-HSCs relatively decreases (Figure 8, lower panel and Figure S5).

Our data show that merely reproducing the ratio of LT-HSCs to ST-HSCs observed in aged mice using young LT-HSCs and ST-HSCs can replicate myeloidbiased hematopoiesis. This suggests that the increase in LT-HSC and the relative decrease in ST-HSC within the HSC compartment with aging are likely to contribute to myeloid-biased hematopoiesis.

As mentioned earlier, since the differentiation capacity of LT-HSCs remain unchanged with age, it seems more accurate to describe that the relative decrease in the proportion of ST-HSCs, which retain long-lived memory lymphocytes in peripheral blood, leads to a relative increase in myeloid cells in peripheral blood and thus causes myeloid-biased hematopoiesis.

However, focusing on the increase in the proportion of LT-HSCs, it is also possible to explain that "with aging, the proportion of LT-HSCs capable of long-term myeloid hematopoiesis increases. As a result, from 16 weeks after transplantation, the influence of LT-HSCs

maintaining the long-term ability to produce myeloid cells becomes relatively more significant, leading to an increase in the ratio of myeloid cells in the peripheral blood and causing myeloid-biased hematopoiesis.”

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

Summary:

Comment #2-1: While aspects of their work are fascinating and might have merit, several issues weaken the overall strength of the arguments and interpretation. Multiple experiments were done with a very low number of recipient mice, showed very large standard deviations, and had no statistically detectable difference between experimental groups. While the authors conclude that these experimental groups are not different, the displayed results seem too variable to conclude anything with certainty. The sensitivity of the performed experiments (e.g. Figure 3; Figure 6C, D) is too low to detect even reasonably strong differences between experimental groups and is thus inadequate to support the author's claims. This weakness of the study is not acknowledged in the text and is also not discussed. To support their conclusions the authors, need to provide higher n-numbers and provide a detailed power analysis of the transplants in the methods section.

Response #2-1

Thank you for your important remarks. The power analysis for this experiment shows that power = 0.319, suggesting that more number may be needed. On the other hand, our method for determining the sample size in Figure 3 is as follows:

- (1) First, we checked whether myeloid biased change is detected in the bulk-HSC fraction (Figure S3). The results showed that the difference in myeloid output at 16 weeks after transplantation was statistically significant (young vs. aged = 7.2 {plus minus} 8.9 vs. 42.1 {plus minus} 35.5%, $p = 0.01$), even though $n = 10$.
- (2) Next, myeloid biased HSCs have been reported to be a fraction with high selfrenewal ability (2004, Blood). If myeloid biased HSCs increase with aging, the increase in myeloid biased HSCs in LT-HSC fraction would be detected with higher sensitivity than in the bulk-HSC fraction used in Figure S3.
- (3) However, there was no difference not only in p-values but also in the mean itself, young vs aged = 51.4{plus minus}31.5% vs 47.4{plus minus}39.0%, $p = 0.82$, even though $n = 8$ in Figure 3. Since there was no difference in the mean itself, it is highly likely that no difference will be detected even if n is further increased.

Regarding Figure 6, we obtained a statistically significant difference and consider the sample size to be sufficient. In addition, we have performed various functional experiments (Figures 2, 5, 6 and S6), and have obtained consistent results that expansion of myeloid-biased HSCs does not occur with aging in Hoxb5+HSCs fraction. Based on the above, we conclude that the LT-HSC fraction does not differ in myeloid differentiation potential with aging.

[Comment for authors]

Paradigm-shifting extraordinary claims require extraordinary data. Unfortunately, the authors do not provide additional data to further support their claims. Instead, the authors argue the following: Because they were able to find significant differences between experimental groups in some experiments, the absence of significant differences in the results of other experiments must be correct, too.

This logic is in my view flawed. Any assay/experiment with highly variable data has a very low sensitivity to detect significant differences between groups. If, as in this case, the variance is as large as the entire dynamic range of the readout, it becomes impossible to be able to detect any difference. In these cases, it is not surprising and actually expected that the mean of the group is located close to the center of the dynamic range as is the case here (center of dynamic range: 50%). In other words, this means that the experiments are simply not reproducible. It is absolutely critical to remember that any experiment and its associated statistical analysis has 3 (!!!) instead of 2 possible outcomes:

- (1) There is a statistically significant difference
- (2) There is no statistically significant difference
- (3) The results of the experiment are inconclusive because the replicates are too variable and the results are not reproducible.

While most of us are inclined to think about outcomes (1) or (2), outcome (3) cannot be neglected. While it might be painful to accept, the only way to address concerns about data reproducibility is to provide additional data, improve reproducibility, and lower the power of the analysis to an acceptable level (e.g. able to detect difference of 5-10% between groups).

Without going into the technical details, the example graph from the link below illustrates that with a power 0.319 as stated by the authors, approx. 25 transplants, instead of 8, would be required.

Typically, however, a power of 0.8 is a reasonable value for any power analysis (although it's not a very strong power either). Even if we are optimistic and assume that there might be a reasonably large difference between experimental groups (in the example above $P_2 = 0.6$, which is actually not that large) we can estimate that we would need over 10 transplants per group to say with confidence that two experimental groups likely do not differ. With smaller differences, these numbers increase quickly to 20+ transplants per group as can be seen in the example graph using an Alpha of 0.1 above.

Further reading can be found here and in many textbooks or other online resources: https://power-analysis.com/effect_size.htm <https://tss.awf.poznan.pl/pdf-188978-110207?filename=Using%20power%20analysis%20to.pdf>

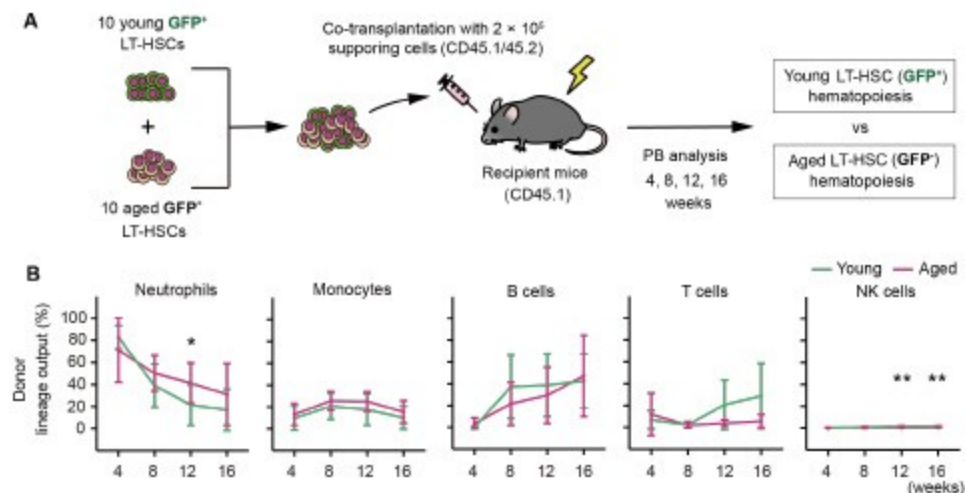
Response:

Thank you for your feedback. We fully agree with the reviewer that paradigmshifting claims must be supported by equally robust data. It has been well documented that the frequency of myeloid-biased HSCs increases with age, with reports indicating that over 50% of the HSC compartment in aged mice consists of myeloid-biased HSCs[1,2]. Based on this, we believe that if aged LT-HSCs were substantially myeloid-biased, the difference should be readily detectable.

To further validate our findings, we showed the similar preliminary experiment. The resulting data are shown below ($n = 8$).

Author response image 1.

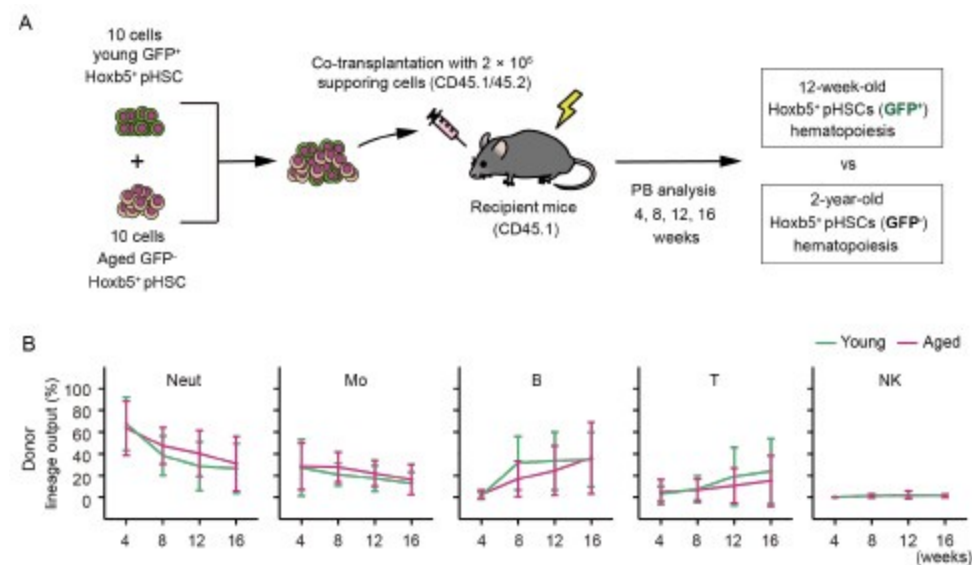
(A) Experimental design for competitive co-transplantation assay. Ten CD45.2⁺ young LT-HSCs and ten CD45.2⁺ aged LT-HSCs were transplanted with 2×10^5 CD45.1⁺/CD45.2⁺ supporting cells into lethally irradiated CD45.1⁺ recipient mice ($n = 8$). (B) Lineage output of young or aged LT-HSCs at 4, 8, 12, 16 weeks after transplantation. Each bar represents an individual mouse. * $P < 0.05$. ** $P < 0.01$.



While a slight increase in myeloid-biased hematopoiesis was observed in the aged LT-HSC fraction, the difference was not statistically significant. These new results are presented alongside the original Figure 3, which was generated using a larger sample size ($n = 16$).

Author response image 2.

(A) Experimental design for competitive co-transplantation assay. Ten CD45.2⁺ young LT-HSCs and ten CD45.2⁺ aged LT-HSCs were transplanted with 2×10^5 CD45.1⁺/CD45.2⁺ supporting cells into lethally irradiated CD45.1⁺ recipient mice ($n = 16$). (B) Lineage output of young or aged LT-HSCs at 4, 8, 12, 16 weeks after transplantation. Each bar represents an individual mouse.



Consistent with the original data, aged LT-HSCs exhibited a lineage output that was nearly identical to that of young LT-HSCs. Nonetheless, as the reviewer rightly pointed out, we cannot completely exclude the possibility that subtle differences may exist but remain undetected. To address this, we have added the following sentence to the manuscript:

[P9, L200] “These findings unmistakably demonstrated that mixed/bulk-HSCs showed myeloid skewed hematopoiesis in PB with aging. In contrast, LT-HSCs maintained a consistent lineage

output throughout life, although subtle differences between aged and young LT-HSCs may exist and cannot be entirely ruled out.”

References

(1) Dykstra, Brad et al. “Clonal analysis reveals multiple functional defects of aged murine hematopoietic stem cells.” *The Journal of experimental medicine* vol. 208,13 (2011): 2691-703. doi:10.1084/jem.20111490

(2) Beerman, Isabel et al. “Functionally distinct hematopoietic stem cells modulate hematopoietic lineage potential during aging by a mechanism of clonal expansion.” *Proceedings of the National Academy of Sciences of the United States of America* vol. 107,12 (2010): 5465-70. doi:10.1073/pnas.1000834107

Comment #2-3: It is also unclear why the authors believe that the observed reduction of LTHSCs relative to LT-HSCs explains the myeloid-biased phenotype observed in the peripheral blood. This point seems counterintuitive and requires further explanation.

Response #2-3:

Thank you for your comment. We apologize for the insufficient explanation. Our data, as shown in Figures 3 and 4, demonstrate that the differentiation potential of LTHSCs remains unchanged with age. Therefore, rather than suggesting that an increase in LT-HSCs with a consistent differentiation capacity leads to myeloid biased hematopoiesis, it seems more accurate to highlight that the relative decrease in the proportion of ST-HSCs, which remain in peripheral blood as lymphocytes, leads to a relative increase in myeloid cells in peripheral blood and thus causes myeloid-biased hematopoiesis. However, if we focus on the increase in the ratio of LT-HSCs, it is also plausible to explain that “with aging, the proportion of LT-HSCs capable of long-term myeloid hematopoiesis increases. As a result, from 16 weeks after transplantation, the influence of LT-HSCs maintaining the long-term ability to produce myeloid cells becomes relatively more significant, leading to an increase in the ratio of myeloid cells in the peripheral blood and causing myeloid-biased hematopoiesis.”

[Comment for authors]

While this interpretation of the data might make sense the shown data do not exclude alternative explanations. The authors do not exclude the possibility that LTHSCs expand with age and that this expansion in combination with an aging microenvironment drives myeloid bias. The authors should quantify the frequency [%] and absolute number of LT-HSCs and ST-HSCs in young vs. aged animals. Especially analyzing the abs. numbers of cells will be important to support their claims as % can be affected by changes in the frequency of other populations.

Thank you for your very important point. As this reviewer pointed out, we do not exclude the possibility that the combination of aged microenvironment drives myeloid bias. Additionally, we acknowledge that myeloid-biased hematopoiesis with age is a complex process likely influenced by multiple factors. We would like to discuss the mechanism mentioned as a future research direction. Thank you for the insightful feedback. Regarding the point about the absolute cell numbers mentioned in the latter half of the paragraph, we will address this in detail in our subsequent response (Response #2-4).

Comment #2-4: Based on my understanding of the presented data, the authors argue that myeloid-biased HSCs do not exist, as a) they detect no difference between young/aged HSCs after transplant (mind low n-numbers and large std!); b) myeloid progenitors downstream of HSCs only show minor or no changes in frequency and c) aged LT-HSCs do not outperform young LT-HSCs in myeloid output LTHSCs in competitive

transplants (mind low n-numbers and large std!). However, given the low n-numbers and high variance of the results, the argument seems weak and the presented data does not support the claims sufficiently. That the number of downstream progenitors does not change could be explained by other mechanisms, for instance, the frequently reported differentiation short-cuts of HSCs and/or changes in the microenvironment.

Response #2-4:

We appreciate the comments. As mentioned above, we will correct the manuscript regarding the sample size. Regarding the interpreting of the lack of increase in the percentage of myeloid progenitor cells in the bone marrow with age, it is instead possible that various confounding factors, such as differentiation shortcuts or changes in the microenvironment, are involved. However, even when aged LT-HSCs and young LT-HSCs are transplanted into the same recipient mice, the timing of the appearance of different cell fractions in peripheral blood is similar (Figure 3 of this paper). Therefore, we have not obtained data suggesting that clear shortcuts exist in the differentiation process of aged HSCs into neutrophils or monocytes. Additionally, it is currently consensually accepted that myeloid cells, including neutrophils and monocytes, differentiate from GMPs¹. Since there are no changes in the proportion of GMPs in the bone marrow with age, we concluded that the differentiation potential into myeloid cells remains consistent with aging.

Reference

(1) Akashi K and others, 'A Clonogenic Common Myeloid Progenitor That Gives Rise to All Myeloid Lineages', *Nature*, 404.6774 (2000), 193-97.

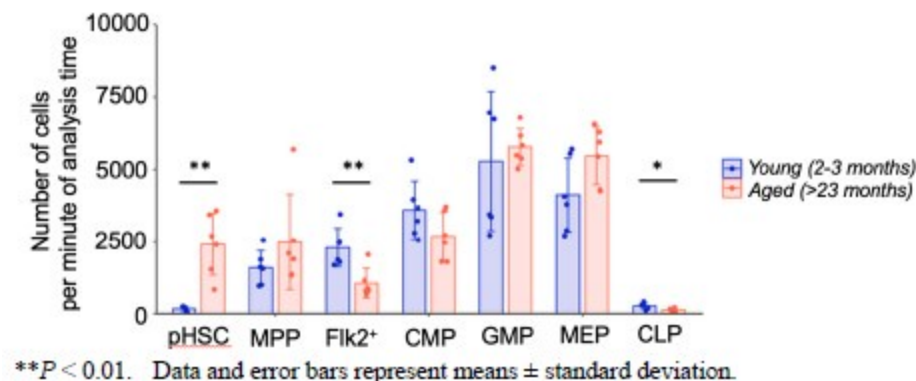
[Comment for authors]

As the relative frequency of cell population can be misleading, the authors should compare the absolute numbers of progenitors in young vs. aged mice to strengthen their argument. It would also be helpful to quantify the absolute numbers and relative frequencies in WT mice to exclude the possibility the HoxB5-trimcherry mouse model suffers from unexpected aging phenotypes and the hematopoietic system differs from wild-type animals.

Thank you for your valuable feedback. We understand the importance of comparing the absolute numbers of progenitors in young versus aged mice to provide a more accurate representation of the changes in cell populations.

Therefore, we quantified the absolute cell count of hematopoietic cells in the bone marrow using flow cytometry data.

Author response image 3.



As previously reported, we observed a 10-fold increase in the number of pHSCs in aged mice compared to young mice. Additionally, our analysis revealed a statistically significant decrease in the number of Flk2+ progenitors and CLPs in aged mice. On the other hand, there was no statistically significant change in the number of myeloid progenitors between the two age groups. We appreciate the suggestion and hope that this additional information strengthens our argument and addresses your concerns.

Comment #2-5:

"Then, we found that the myeloid lineage proportions from young and aged LT-HSCs were nearly comparable during the observation period after transplantation (Figure 3, B and C)." Given the large standard deviation and low n-numbers, the power of the analysis to detect differences between experimental groups is very low. Experimental groups with too large standard deviations (as displayed here) are difficult to interpret and might be inconclusive. The absence of clearly detectable differences between young and aged transplanted HSCs could thus simply be a false-negative result. The shown experimental results hence do not provide strong evidence for the author's interpretation of the data. The authors should add additional transplants and include a detailed power analysis to be able to detect differences between experimental groups with reasonable sensitivity.

Response #2-5:

Thank you for providing these insights. Regarding the sample size, we have addressed this in Response #2-1.

[Comment for authors]

As explained in detail in the response to #2-1 the provided arguments are not convincing. As the authors pointed out, the power of these experiments is too low to make strong claims. If the author does not intend to provide new data, the language of the manuscript needs to be adjusted to reflect this weakness. A paragraph discussing the limitations of the study mentioning the limited power of the data should be included beyond the above-mentioned rather vague statement that the data should be validated (which is almost always necessary anyway).

Thank you for your valuable comment. We agree with the importance of discussing potential limitations in our experimental design. In response to the reviewer's suggestion, we have

revised the manuscript to include the following sentences:

[P19, L434] "In the co-transplantation assay shown in Figure 3, the myeloid lineage output derived from young and aged LT-HSCs was comparable (Young LT-HSC: $51.4 \pm 31.5\%$ vs. Aged LT-HSC: $47.4 \pm 39.0\%$, $p = 0.82$). Although no significant difference was detected, the small sample size ($n = 8$) may limit the sensitivity of the assay to detect subtle myeloid-biased phenotypes."

This addition acknowledges the potential limitations of our analysis and highlights the need for further investigation with larger cohorts.

Comment #2-6:

Line 293: "Based on these findings, we concluded that myeloid biased hematopoiesis observed following transplantation of aged HSCs was caused by a relative decrease in ST-HSC in the bulk-HSC compartment in aged mice rather than the selective expansion of myeloid-biased HSC clones." Couldn't that also be explained by an increase in myeloid-biased HSCs, as repeatedly reported and seen in the expansion of CD150+ HSCs? It is not intuitively clear why a reduction of STHSCs clones would lead to a myeloid bias. The author should try to explain more clearly where they believe the increased number of myeloid cells comes from. What is the source of myeloid cells if the authors believe they are not derived from the expanded population of myeloid-biased HSCs?

Response #2-6:

Thank you for pointing this out. We apologize for the insufficient explanation. We will explain using attached Figure 8 from the paper. First, our data show that LT-HSCs maintain their differentiation capacity with age, while ST-HSCs lose their self-renewal capacity earlier, so that only long-lived memory lymphocytes remain in the peripheral blood after the loss of self-renewal capacity in ST-HSCs (Figure 8, upper panel). In mouse bone marrow, the proportion of LT-HSCs increases with age, while the proportion of STHSCs relatively decreases (Figure 8, lower panel and Figure S5).

Our data show that merely reproducing the ratio of LT-HSCs to ST-HSCs observed in aged mice using young LT-HSCs and ST-HSCs can replicate myeloid-biased hematopoiesis. This suggests that the increase in LT-HSC and the relative decrease in ST-HSC within the HSC compartment with aging are likely to contribute to myeloid-biased hematopoiesis.

As mentioned earlier, since the differentiation capacity of LT-HSCs remain unchanged with age, it seems more accurate to describe that the relative decrease in the proportion of STHSCs, which retain long-lived memory lymphocytes in peripheral blood, leading to a relative increase in myeloid cells in peripheral blood and thus causes myeloid-biased hematopoiesis. However, focusing on the increase in the proportion of LT-HSCs, it is also possible to explain that "with aging, the proportion of LT-HSCs capable of long-term myeloid hematopoiesis increases. As a result, from 16 weeks after transplantation, the influence of LT-HSCs maintaining the long-term ability to produce myeloid cells become relatively more significant, leading to an increase in the ratio of myeloid cells in the peripheral blood and causing myeloid biased hematopoiesis."

[Comment for authors]

While I can follow the logic of the argument, my concerns about the interpretation remain as I see discrepancies in other findings in the published literature. For instance, what the authors call ST-HSCs, differs from the classical functional definition of ST-HSCs. It is thus difficult to relate the described observations to previous reports. ST-HSCs typically can contribute significantly to multiple lineages for several weeks (see for

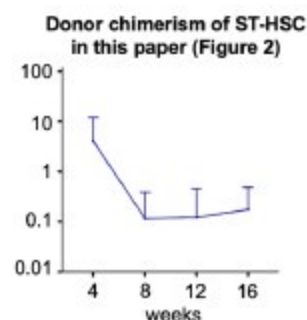
example PMID: 29625072). It is somewhat surprising that the ST-HSC in this study don't show this potential and lose their potential much quicker.

The authors should thus provide a more comprehensive depth of immunophenotypic and molecular characterization to compare their LT-HSCs to ST-HSCs. For instance, are LT-HSCs CD41⁺ HSCs? How do ST-HSCs differ in their surface marker expression from previously used definitions of ST-HSCs? A list of differentially expressed genes between young and old LT-HSCs and ST-HSCs should be done and will likely provide important insights into the molecular programs/markers (beyond the provided GO analysis, which seems superficial).

Thank you for your valuable feedback. As the reviewer noted, there are indeed multiple definitions of ST-HSCs. We appreciate the opportunity to clarify our definitions of ST-HSCs. We define ST-HSCs functionally, rather than by surface antigens, which we believe is the most classical and widely accepted definition [1]. In our study, we define long-term hematopoietic stem cells (LT-HSCs) as those HSCs that continue to contribute to hematopoiesis after a second transplantation and possess long-term self-renewal potential. Conversely, we define short-term hematopoietic stem cells (ST-HSCs) as those HSCs that do not contribute to hematopoiesis after a second transplantation and only exhibit self-renewal potential in the short term.

Next, in the paper referenced by the reviewer[2], the chimerism of each fraction of ST-HSCs also peaked at 4 weeks and then decreased to approximately 0.1% after 12 weeks post-transplantation. Author response image 5 illustrates our ST-HSC donor chimerism in Figure 2. We believe that data in the paper referenced by the reviewer2 is consistent with our own observations of the hematopoietic pattern following ST-HSC transplantation, indicating a characteristic loss of hematopoietic potential 4 weeks after the transplantation. Furthermore, as shown in Figures 2D and 2F, the fraction of ST-HSCs does not exhibit hematopoietic activity after the second transplantation. Therefore, we consider this fraction to be ST-HSCs.

Author response image 4.



Additionally, the RNAseq data presented in Figures 4 and S4 revealed that the GSEA results vary among the different myeloid gene sets analyzed (Fig. 4, D–F; Fig. S4, C–D). Moreover, a comprehensive analysis of mouse HSC aging using multiple RNA-seq datasets reported that nearly 80% of differentially expressed genes show poor reproducibility across datasets[3]. From the above, while RNAseq data is indeed helpful, we believe that emphasizing functional experimental results is more critical than incorporating an additional dataset to support our claim. Thank you once again for your insightful feedback.

References

- (1) Kiel, Mark J et al. "SLAM family receptors distinguish hematopoietic stem and progenitor cells and reveal endothelial niches for stem cells." *Cell* vol. 121,7 (2005): 1109-21.

doi:10.1016/j.cell.2005.05.026

(2) Yamamoto, Ryo et al. "Large-Scale Clonal Analysis Resolves Aging of the Mouse Hematopoietic Stem Cell Compartment." *Cell stem cell* vol. 22,4 (2018): 600-607.e4.
doi:10.1016/j.stem.2018.03.013

(3) Flohr Svendsen, Arthur et al. "A comprehensive transcriptome signature of murine hematopoietic stem cell aging." *Blood* vol. 138,6 (2021): 439-451.
doi:10.1182/blood.2020009729

Reviewer #3 (Public review):

Although the topic is appropriate and the new model provides a new way to think about lineage-biased output observed in multiple hematopoietic contexts, some of the experimental design choices, as well as some of the conclusions drawn from the results could be substantially improved. Also, they do not propose any potential mechanism to explain this process, which reduces the potential impact and novelty of the study.

The authors have satisfactorily replied to some of my comments. However, there are multiple key aspects that still remain unresolved.

Reviewer #3 (Recommendations for the authors):

Comment #3-1,2:

Although the additional details are much appreciated the core of my original comments remains unanswered. There are still no details about the irradiation dose for each particular experiment. Is any transplant performed using a 9.1 Gy dose? If yes, please indicate it in text or figure legend. If not, please remove this number from the corresponding method section.

Again, 9.5 Gy (split in two doses) is commonly reported as sublethal. The fact that the authors used a methodology that deviates from the "standard" for the field makes difficult to put these results in context with previous studies. It is not possible to know if the direct and indirect effects of this conditioning method in the hematopoietic system have any consequences in the presented results.

Thank you for your clarification. We confirm that none of the transplantation experiments described were performed using a 9.1 Gy irradiation dose. We have therefore removed the mention of "9.1 Gy" from the relevant section of the Materials and Methods. We appreciate helpful suggestion to improve the clarity of the manuscript.

[P22, L493] "12-24 hours prior to transplantation, C57BL/6-Ly5.1 mice, or aged C57BL/6J recipient mice were lethally irradiated with single doses of 8.7 Gy."

Regarding the reviewer's concern about the radiation dose used in our experiments, we will address this point in more detail in our subsequent response (see Response #3-4).

Comment #3-4(Original): When representing the contribution to PB from transplanted cells, the authors show the % of each lineage within the donor-derived cells (Figures 3B-C, 5B, 6B-D, 7C-E, and S3 B-C). To have a better picture of total donor contribution, total PB and BM chimerism should be included for each transplantation assay. Also, for Figures 2C-D and Figures S2A-B, do the graphs represent 100% of the PB cells? Are there any radioresistant cells?

Response #3-4 (Original): Thank you for highlighting this point. Indeed, donor contribution to total peripheral blood (PB) is important information. We have included

the donor contribution data for each figure above mentioned.

In Figure 2C-D and Figure S2A-B, the percentage of donor chimerism in PB was defined as the percentage of CD45.1-CD45.2+ cells among total CD45.1-CD45.2+ and CD45.1+CD45.2+ cells as described in method section.

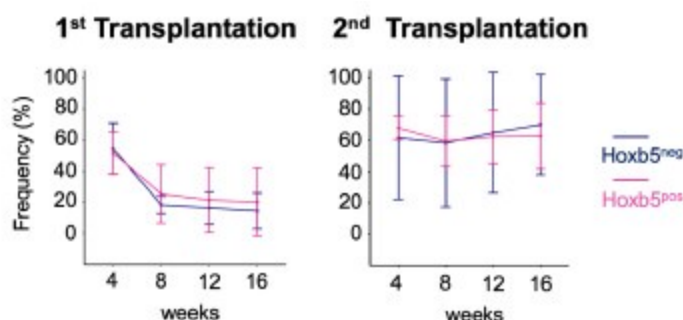
Comment for our #3-4 response:

Thanks for sharing these data. These graphs should be included in their corresponding figures along with donor contribution to BM.

Regarding Figure 2 C-D, as currently shown, the graphs only account for CD45.1CD45.2+ (donor-derived) and CD45.1+CD45.2+ (supporting-derived). What is the percentage of CD45.1+CD45.2- (recipient-derived)? Since the irradiation regimen is atypical, including this information would help to know more about the effects of this conditioning method.

Thank you for your insightful comment regarding Figure 2C-D. To address the concern that the reviewer pointed out, we provide the kinetics of the percentage of CD45.1+CD45.2- (recipient-derived) in Author response image 7.

Author response image 5.



As the reviewer pointed out, we observed the persistence of recipient-derived cells, particularly in the secondary transplant. As noted, this suggests that our conditioning regimen may have been suboptimal. In response, we will include the donor chimerism analysis in the total cells and add the following statement in the study limitations section to acknowledge this point:

[P19, L439] “Additionally, in this study, we purified LT-HSCs using the Hoxb5 reporter system and employed a moderate conditioning regimen (8.7 Gy). To have a better picture of total donor contribution, total PB chimerism are presented in Figure S7 and we cannot exclude the possibility that these factors may have influenced the results. Therefore, it would be ideal to validate our findings using alternative LT-HSC markers and different conditioning regimens.”

Comment #3-5: For BM progenitor frequencies, the authors present the data as the frequency of cKit+ cells. This normalization might be misleading as changes in the proportion of cKit+ between the different experimental conditions could mask differences in these BM subpopulations. Representing this data as the frequency of BM single cells or as absolute numbers (e.g., per femur) would be valuable.

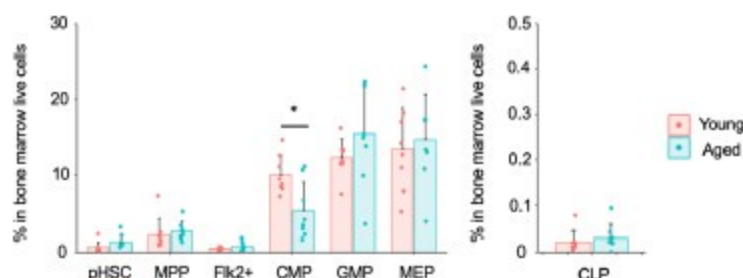
Response #3-5:

We appreciate the reviewer's comment on this point.

Firstly, as shown in Supplemental Figures S1B and S1C, we analyze the upstream (HSC, MPP, Flk2+) and downstream (CLP, MEP, CMP, GMP) fractions in different panels. Therefore, normalization is required to assess the differentiation of HSCs from upstream to downstream.

Additionally, the reason for normalizing by c-Kit+ is that the bone marrow analysis was performed after enrichment using the Anti-c-Kit antibody for both upstream and downstream fractions. Based on this, we calculated the progenitor populations as a frequency within the c-Kit positive cells. Next, the results of normalizing the whole bone marrow cells (live cells) are shown below.

Author response image 6.



Similar to the results of normalizing c-Kit+ cells, myeloid progenitors remained unchanged, including a statistically significant decrease in CMP in aged mice. Additionally, there were no significant differences in CLP. In conclusion, similar results were obtained between the normalization with c-Kit and the normalization with whole bone marrow cells (live cells).

However, as the reviewer pointed out, it is necessary to explain the reason for normalization with c-Kit. Therefore, we will add the following description.

[P21, L502] For the combined analysis of the upstream (HSC, MPP, Flk2+) and downstream (CLP, MEP, CMP, GMP) fractions in Figures 1B, we normalized by cKit+ cells because we performed a c-Kit enrichment for the bone marrow analysis.

Comment for our #3-5 response:

I understand that normalization is necessary to compare across different BM populations. However, the best way would be to normalize to single cells. As I mentioned in my original comment, normalizing to cKit+ cells could be misleading, as the proportion of cKit+ cells could be different across the experimental conditions. Further, enriching for cKit+ cells when analyzing BM subpopulation frequencies could introduce similar potential errors. The enrichment would depend on the level of expression of cKit for each of these population, what would alter the final quantification. Indeed, CLP are typically defined as cKit-med/low. Thus, cKit enrichment would not be a great method to analyze the frequency of these cells.

The graph in the authors' response to my comment, show similar trend to what is represented Figure 1B for some populations. However, there are multiple statistically significant changes that disappear in this new version. This supports my original concern and, in consequence, I would encourage to represent this data as the frequency of BM single cells or as absolute numbers (e.g., per femur).

Thank you for your thoughtful follow-up comment. In response to the reviewer's suggestion, we will represent the data as the frequency among total BM single cells. These revised graphs have been incorporated into the updated Figure 7F and corresponding figure legend have

been revised accordingly to accurately reflect these representations. We appreciate your valuable input, which has helped us improve the clarity and rigor of our data presentation.

Comment #3-6: Regarding Figure 1B, the authors argue that if myeloid-biased HSC clones increase with age, they should see increased frequency of all components of the myeloid differentiation pathway (CMP, GMP, MEP). This would imply that their results (no changes or reduction in these myeloid subpopulations) suggest the absence of myeloid-biased HSC clones expansion with age. This reviewer believes that differentiation dynamics within the hematopoietic hierarchy can be more complex than a cascade of sequential and compartmentalized events (e.g., accelerated differentiation at the CMP level could cause exhaustion of this compartment and explain its reduction with age and why GMP and MEP are unchanged) and these conclusions should be considered more carefully.

Response #3-6:

We wish to thank the reviewer for this comment. We agree with that the differentiation pathway may not be a cascade of sequential events but could be influenced by various factors such as extrinsic factors.

In Figure 1B, we hypothesized that there may be other mechanisms causing myeloid-biased hematopoiesis besides the age-related increase in myeloid-biased HSCs, given that the percentage of myeloid progenitor cells in the bone marrow did not change with age. However, we do not discuss the presence or absence of myeloid-biased HSCs based on the data in Figure 1B.

Our newly proposed theories—that the differentiation capacity of LT-HSCs remains unchanged with age and that age-related myeloid-biased hematopoiesis is due to changes in the ratio of LT-HSCs to ST-HSCs—are based on functional experiment results. As the reviewer pointed out, to discuss the presence or absence of myeloid-biased HSCs based on the data in Figure 1B, it is necessary to apply a system that can track HSC differentiation at single-cell level. The technology would clarify changes in the self-renewal capacity of individual HSCs and their differentiation into progenitor cells and peripheral blood cells. The authors believe that those single-cell technologies will be beneficial in understanding the differentiation of HSCs. Based on the above, the following statement has been added to the text.

[P19, L440] In contrast, our findings should be considered in light of some limitations. In this report, we primarily performed ten to twenty cell transplantation assays. Therefore, the current theory should be revalidated using single-cell technology with lineage tracing system1-2. This approach will investigate changes in the self-renewal capacity of individual HSCs and their subsequent differentiation into progenitor cells and peripheral blood cells.

Comment for our #3-6 response:

Thanks for the response. My original comments referred to the statement "On the other hand, in contrast to what we anticipated, the frequency of GMP was stable, and the percentage of CMP actually decreased significantly with age, defying our prediction that the frequency of components of the myeloid differentiation pathway, such as CMP, GMP, and MEP would increase in aged mice if myeloid-biased HSC clones increase with age (Fig. 1 B)" (lines #129-133). Again, the absence of an increase in CMP, GMP and MEP with age does not mean the absence of and increase in myeloid-biased HSC clones. This statement should be considered more carefully.

Thank you for the insightful comment. We agree that the absence of an increase in CMP, GMP and MEP with age does not mean the absence of an increase in myeloid-biased HSC clones. In

our revised manuscript, we have refined the statement to acknowledge this nuance more clearly. The updated text now reads as follows:

P6, L129] On the other hand, in contrast to what we anticipated, the frequency of GMP was stable, and the percentage of CMP actually decreased significantly with age, defying our prediction that the frequency of components of the myeloid differentiation pathway, such as CMP, GMP, and MEP may increase in aged mice, if myeloid-biased HSC clones increase with age.

Comment #3-7: Within the few recipients showing good donor engraftment in Figure 2C, there is a big proportion of T cells that are "amplified" upon secondary transplantation (Figure 2D). Is this expected?

Response #3-7:

We wish to express our deep appreciation to the reviewer for insightful comment on this point. As the reviewers pointed out, in Figure 2D, a few recipients show a very high percentage of T cells. The authors had the same question and considered this phenomenon as follows:

(1) One reason for the very high percentage of T cells is that we used 1×10^7 whole bone marrow cells in the secondary transplantation. Consequently, the donor cells in the secondary transplantation contained more T-cell progenitor cells, leading to a greater increase in T cells compared to the primary transplantation.

(2) We also consider that this phenomenon may be influenced by the reduced selfrenewal capacity of aged LT-HSCs, resulting in decreased sustained production of myeloid cells in the secondary recipient mice. As a result, long-lived memorytype lymphocytes may preferentially remain in the peripheral blood, increasing the percentage of T cells in the secondary recipient mice.

We have discussed our hypothesis regarding this interesting phenomenon. To further clarify the characteristics of the increased T-cell count in the secondary recipient mice, we will analyze TCR clonality and diversity in the future.

Comment for our #3-7 response:

Thanks for the potential explanations to my question. This fact is not commonly reported in previous transplantation studies using aged HSCs. Could Hoxb5 label fraction of HSCs that is lymphoid/T-cell biased upon secondary transplantation? The number of recipients with high frequency of lymphoid cells in the peripheral blood (even from young mice) is remarkable.

Response:

Thank you for your insightful suggestion. Based on this comment, we calculated the percentage of lymphoid cells in the donor fraction at 16 weeks following the secondary transplantation, which was $56.1 \pm 25.8\%$ (L/M = 1.27). According to the Müller-Sieburg criteria, lymphoid-biased hematopoiesis is defined as having an L/M ratio greater than 10.

Given our findings, we concluded that the Hoxb5-labeled fraction does not specifically indicate lymphoid-biased hematopoiesis. We sincerely appreciate the valuable input, which helped us to further clarify the interpretation of our results.

Comment #3-8: Do the authors have any explanation for the high level of variability within the recipients of Hoxb5+ cells in Figure 2C?

Response #3-8:

We appreciate the reviewer's comment on this point. As noted in our previous report, transplantation of a sufficient number of HSCs results in stable donor chimerism, whereas a small number of HSCs leads to increased variability in donor chimerism¹. Additionally, other studies have observed high variability when fewer than 10 HSCs are transplanted²⁻³. Based on this evidence, we consider that the transplantation of a small number of cells (10 cells) is the primary cause of the high level of variability observed.

Comment for our #3-8 response:

I agree that transplanting low number of HSC increases the mouse-to-mouse variability. For that reason, a larger cohort of recipients for this kind of experiment would be ideal.

Response:

Thank you for the insightful comment. We agree that a larger cohort of recipients would be ideal for this type of experiment. In Figure 2, the difference between Hoxb5⁺ and Hoxb5⁻ cells are robust, allowing for a clear statistical distinction despite the cohort size. However, we also recognize that a larger cohort would be necessary to detect more subtle differences, particularly in Figure 3. In response, we have added the following statement to the main text to acknowledge this limitation.

P9, L200] These findings unmistakably demonstrated that mixed/bulk-HSCs showed myeloid skewed hematopoiesis in PB with aging. In contrast, LT-HSCs maintained a consistent lineage output throughout life, although subtle differences between aged and young LT-HSCs may exist and cannot be entirely ruled out.

Comment #3-10: Is Figure 2G considering all primary recipients or only the ones that were used for secondary transplants? The second option would be a fairer comparison.

Response #3-10:

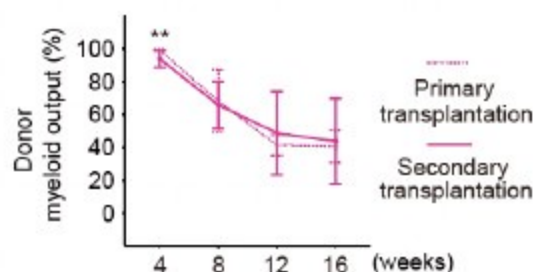
We appreciate the reviewer's comment on this point. We considered all primary recipients in Figure 2G to ensure a fair comparison, given the influence of various factors such as the radiosensitivity of individual recipient mice^[1]. Comparing only the primary recipients used in the secondary transplantation would result in $n = 3$ (primary recipient) vs. $n = 12$ (secondary recipient). Including all primary recipients yields $n = 11$ vs. $n = 12$, providing a more balanced comparison. Therefore, we analyzed all primary recipient mice to ensure the reliability of our results.

Comment for our #3-10 response:

I respectfully disagree. Secondary recipients are derived from only 3 of the primary recipients. Therefore, the BM composition is determined by the composition of their donors. Including primary recipients that are not transplanted into secondary recipients for is not the fairest comparison for this analysis.

Thank you for your comment and for highlighting this important issue. We acknowledge the concern that including primary recipients that are not transplanted into secondary recipients is not the fairest comparison for this analysis. In response, we have reanalyzed the data using only the primary recipients whose bone marrow was actually transplanted into secondary recipients.

Author response image 7.



Importantly, the reanalysis confirmed that the kinetics of myeloid cell proportions in peripheral blood were consistent between primary and secondary transplant recipients. We sincerely appreciate your thoughtful feedback, which has helped us improve the clarity.

Comment #3-11: When discussing the transcriptional profile of young and aged HSCs, the authors claim that genes linked to myeloid differentiation remain unchanged in the LT-HSC fraction while there are significant changes in the STHSCs. However, 2 out of the 4 genes shown in Figure S4B show ratios higher than 1 in LT-HSCs.

Response #3-11:

Thank you for highlighting this important point. As the reviewer pointed out, when we analyze the expression of myeloid-related genes, some genes are elevated in aged LT-HSCs compared to young LT-HSCs. However, the GSEA analysis using myeloid-related gene sets, which include several hundred genes, shows no significant difference between young and aged LT-HSCs (see Figure S4C in this paper). Furthermore, functional experiments using the co-transplantation system show no difference in differentiation capacity between young and aged LT-HSCs (see Figure 3 in this paper). Based on these results, we conclude that LT-HSCs do not exhibit any change in differentiation capacity with aging.

Comment for our #3-11 response:

The authors used the data in Figure S4 to claim that "myeloid genes were tended to be enriched in aged bulk-HSCs but not in aged LT-HSCs compared to their respective controls" (this is the title of the figure; line # 1326). This is based on an increase in gene expression of CD150, vWF, Selp, Itgb3 in aged cells compared to young cells (Figure S4B). However, an increase in Selp and Itgb3 is also observed for LT-HSCs (lower magnitude, but still and increase).

Also, regarding the GSEA, the only term showing statistical significance in bulk HSCs is "Myeloid gene set", which does not reach significance in LT-HSCs, but present a trend for enrichment ($q = 0.077$). None of the terms in shown in this panel present statistical significance in ST-HSCs.

Thank you for your valuable point. As the reviewer noted, the current title may cause confusion. Therefore, we propose changing it to the following:

[P52, L1331] "Figure S4. Compared to their respective young controls, aged bulk-HSCs exhibit greater enrichment of myeloid gene expression than aged LT-HSCs"

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